Mathematics of Machine Learning

Bayesian Inference



Bayes, Conjugate Priors, Uninformative Priors

Outline

1. **Bayes' Rule**
2. Conjugate Priors
3. Uninformative Priors



Dealing with Uncertainty

- In machine learning we are attempting to make inference under uncertainty
- The natural language for discussing uncertainty is probability
- The natural framework for making inferences is Bayesian statistics
- However, this requires that we encode our prior knowledge of the problem and specify a likelihood
- In consequence, probabilistic methods tend to be bespoke, rather than general purpose black boxes

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- Bayes' rule

$$\mathbb{P}(\mathcal{H}_i|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\mathcal{H}_i)\mathbb{P}(\mathcal{H}_i)}{\mathbb{P}(\mathcal{D})}$$

  - $\mathbb{P}(\mathcal{H}_i|\mathcal{D})$ is the **posterior** probability of a hypothesis $\mathcal{H}_i$ (i.e. the probability of $\mathcal{H}_i$ **after** we see the data)
  - $\mathbb{P}(\mathcal{D}|\mathcal{H}_i)$ is the **likelihood** of the data given the hypothesis. Note, that we calculated this from the forward problem
  - $\mathbb{P}(\mathcal{H}_i)$ is the **prior** probability (i.e. the probability of $\mathcal{H}_i$ **before** we see the data)
  - $\mathbb{P}(\mathcal{D})$ is the **evidence**

$$\mathbb{P}(\mathcal{D}) = \sum_{i=1}^{n} \mathbb{P}(\mathcal{H}_i, \mathcal{D}) = \sum_{i=1}^{n} \mathbb{P}(\mathcal{D}|\mathcal{H}_i)\mathbb{P}(\mathcal{H}_i)$$

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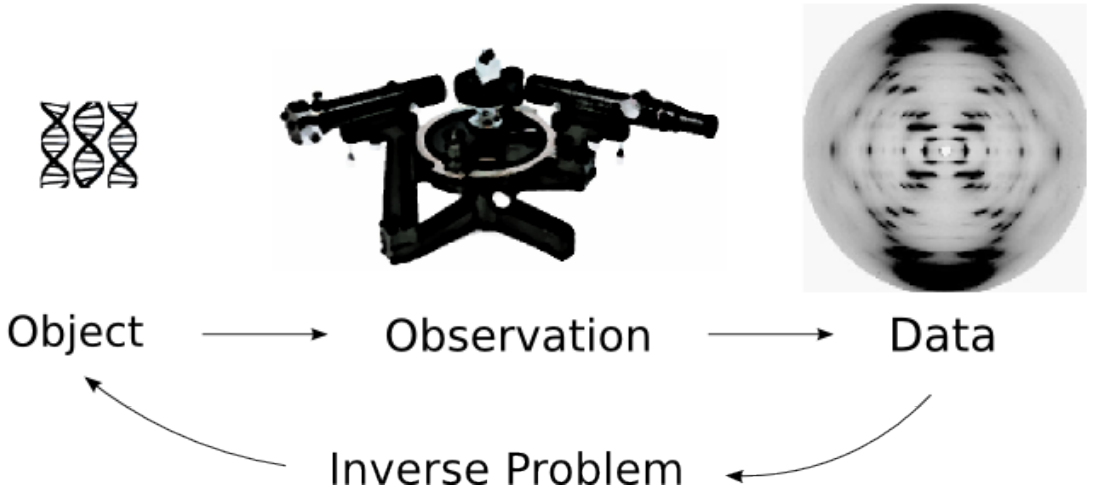
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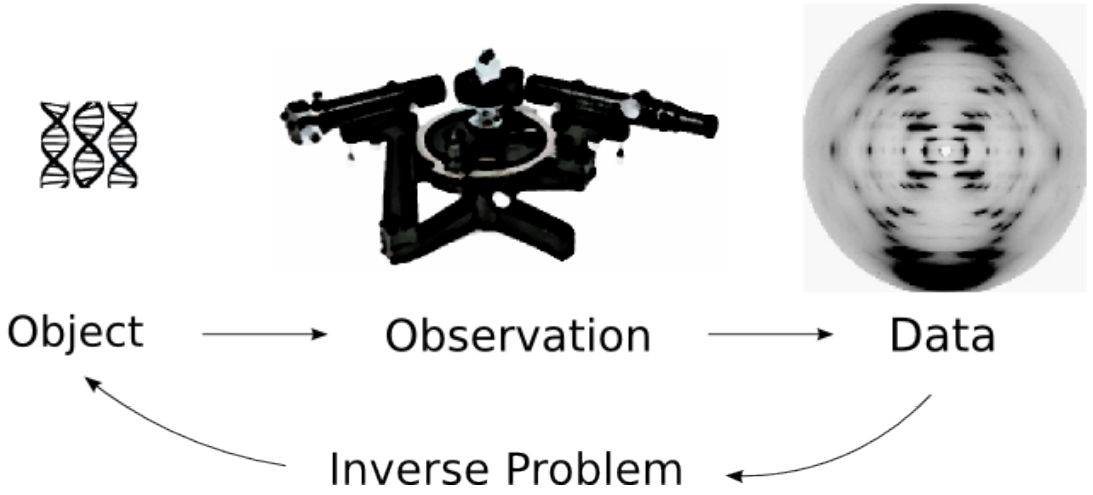
Solving Inverse Problems



- We want the posterior $\mathbb{P}(\mathcal{H}_i|\mathcal{D})$ (i.e. the probability of what happened given some evidence)
- The Bayesian formalism converts this into the forward problem

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- We calculate the likelihood $\mathbb{P}(\mathcal{D}|\mathcal{H}_i)$ (i.e. assuming the hypothesis, what is the chance of obtaining the data?)
- We consider the process of how the data is generated
- This uses the data we have (doesn't care about missing data)
- But we also need to know the prior $\mathbb{P}(\mathcal{H}_i)$
- Also, this can get difficult when we have many hypotheses

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- The normalisation term

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  tells you how likely the data is (given the prior and likelihood function)

- It is called the **marginal likelihood**

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- But it also plays an important role in **model selection**

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Probability Density

- When we are working with continuous variables it is more natural to work with probability densities

$$f_X(x) = \lim_{\delta x \to 0} \frac{\mathbb{P}(x \leq X < x + \delta x)}{\delta x}$$

- Note that densities are non-negative, but can be greater than 1 (they are not probabilities)

- However

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) \mathrm{d}x$$

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Densities and Bayes

- Bayes' rule also applies to densities

$$\mathbb{P}(x \leq X < x + \delta x | Y) = \frac{\mathbb{P}(Y|x)\mathbb{P}(x \leq X < x + \delta x)}{\mathbb{P}(Y)}$$

- Dividing by δx and taking the limit $\delta x \to 0$

$$f_{X|Y}(x|Y) = \frac{\mathbb{P}(Y|x) f_X(x)}{\mathbb{P}(Y)}$$

- Similarly if X is discrete and Y continuous

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Practical Bayesian Inference

- Often consider learning parameters $\boldsymbol{\theta}$

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}$$

- This can be hard for large data sets as the posterior, $p(\boldsymbol{\theta}|\mathcal{D})$, is often a mess
- If we are lucky and have a simple likelihood then if we choose the right prior we end up with a posterior of the same form as the prior
- This occurs in some classic probabilistic inference problems, but as we will see soon it is also true for Gaussian Processes

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Learning a Probability

- Suppose we have a coin and we want to establish the probability of a head
- We want to learn this from a series of independent trials
- (Independent trials with two possible outcomes are known in probability theory as Bernoulli trials)
- Let X_i equal 1 if the ith trial is a head and 0 otherwise
- If the probability of a head is p then the **likelihood** of a X_i is

$$\mathbb{P}(X_i|p) = p^{X_i}(1-p)^{1-X_i}$$

Learning a Probability

- Suppose we have a coin and we want to establish the probability of a head
- We want to learn this from a series of independent trials
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- Let X_i equal 1 if the ith trial is a head and 0 otherwise
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Prior

- We may have a prior belief (e.g. we have made a few trials or we see the coin looks like a normal penny)

- We will suppose we can model our prior belief in terms of a **Beta distribution**

$$f(p) = \mathrm{Beta}(p|a,b) = \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}$$

- $B(a,b)$ is just a normalisation constant

$$B(a,b) = \int_0^1 p^{a-1}(1-p)^{b-1}\mathrm{d}p = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

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- Suppose we have no idea about p what should we do?
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Independent Trials

- Using Bayes' rule

$$f(p|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|p)\, f(p)}{\mathbb{P}(\mathcal{D})}$$

- Assuming the trials are independent (a reasonably fair assumption for tossing coins) then the likelihood factorises

$$\begin{aligned}\mathbb{P}(\mathcal{D}|p) &= \prod_{i=1}^{n} p^{X_i}(1-p)^{1-X_i} \\ &= p^{X_1}(1-p)^{1-X_1} p^{X_2}(1-p)^{1-X_2} \cdots p^{X_n}(1-p)^{1-X_n} \\ &= p^{\sum_i X_i}(1-p)^{\sum_i (1-X_i)} = p^s (1-p)^{n-s}\end{aligned}$$

$s = \sum_i X_i$ (number of successes/heads)

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- The posterior distribution is Beta distribution

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- Starting with a beta distributed prior $f(p) = \mathrm{Beta}(p|a_0, b_0)$ for a set of Bernoulli trials we obtain a beta distributed posterior $f(p|\mathcal{D}) = \mathrm{Beta}(p|a_0 + s, b_0 + n - s)$
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Incremental Updating

- The X_i are independent **given** p, so we can update incrementally, $\mathcal{D} = (X_1, X_2, \ldots, X_n)$

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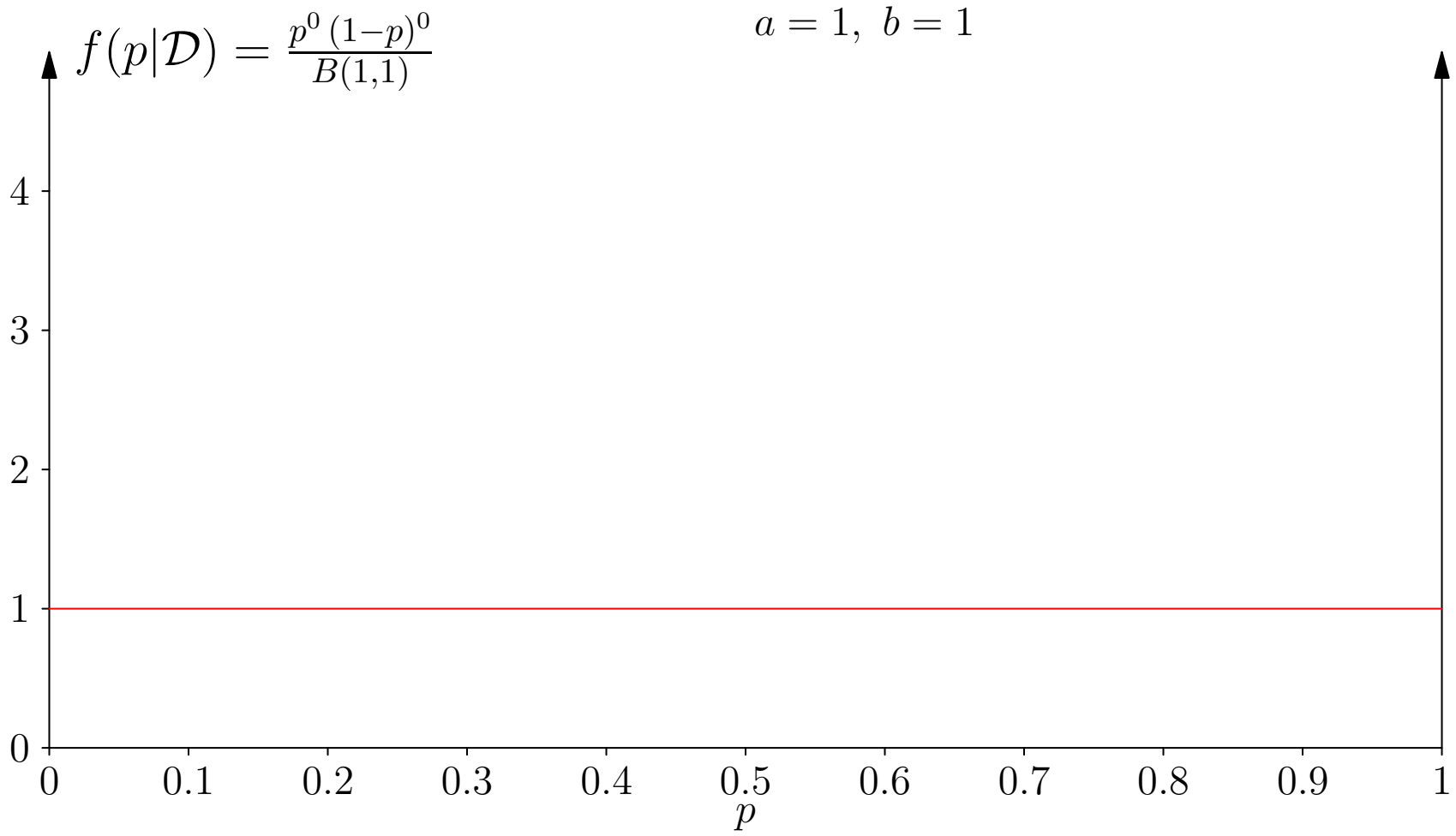
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Example (p=0.7)

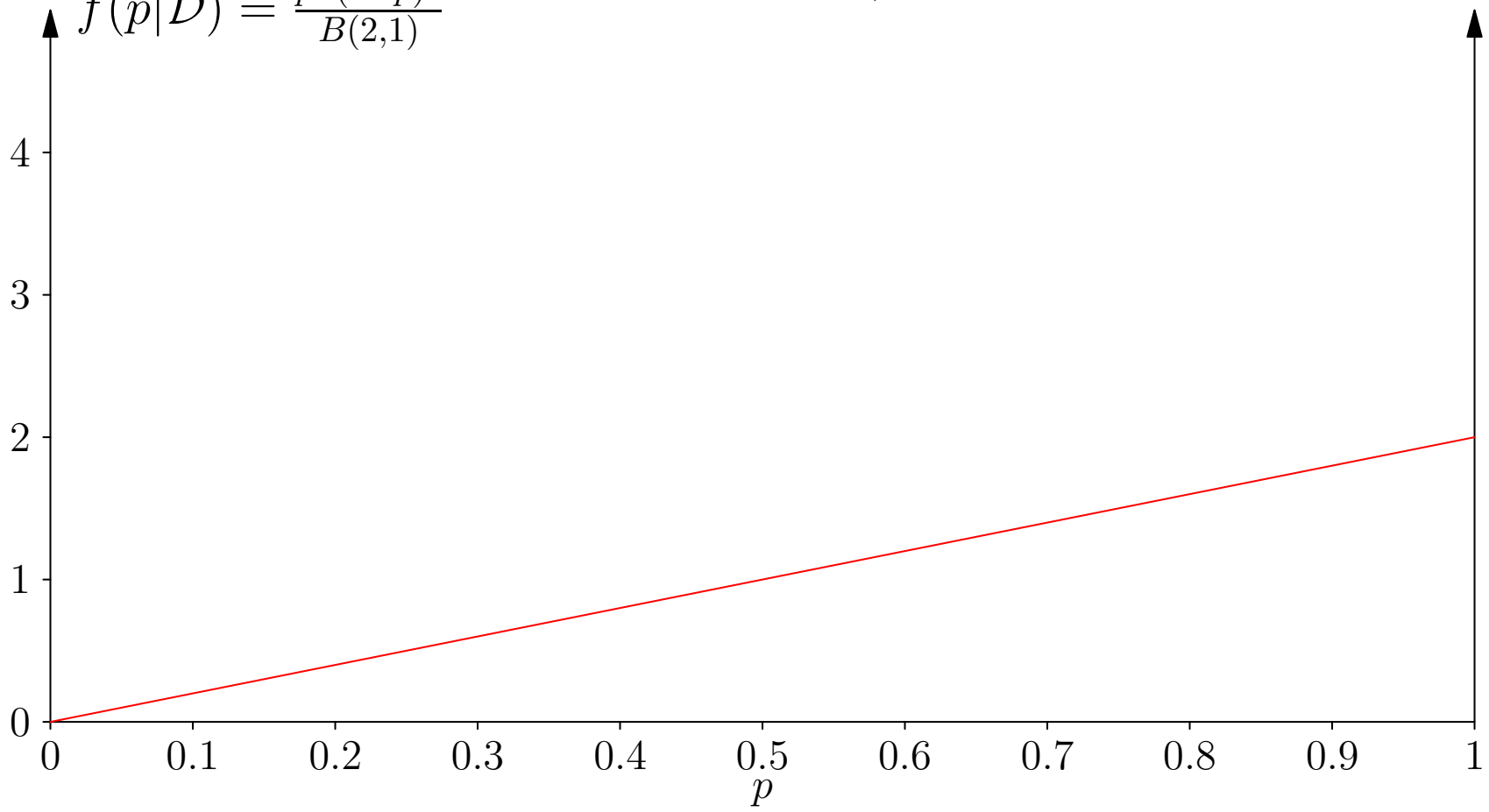

$f(p|\mathcal{D}) = \frac{p^0\,(1-p)^0}{B(1,1)}$
$a = 1,\ b = 1$
4
3
2
1
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
p

Example (p=0.7)

$\mathcal{D} = \{\mathrm{H},$

$f(p|\mathcal{D}) = \frac{p^1 (1-p)^0}{B(2,1)}$

$a = 2,\ b = 1$

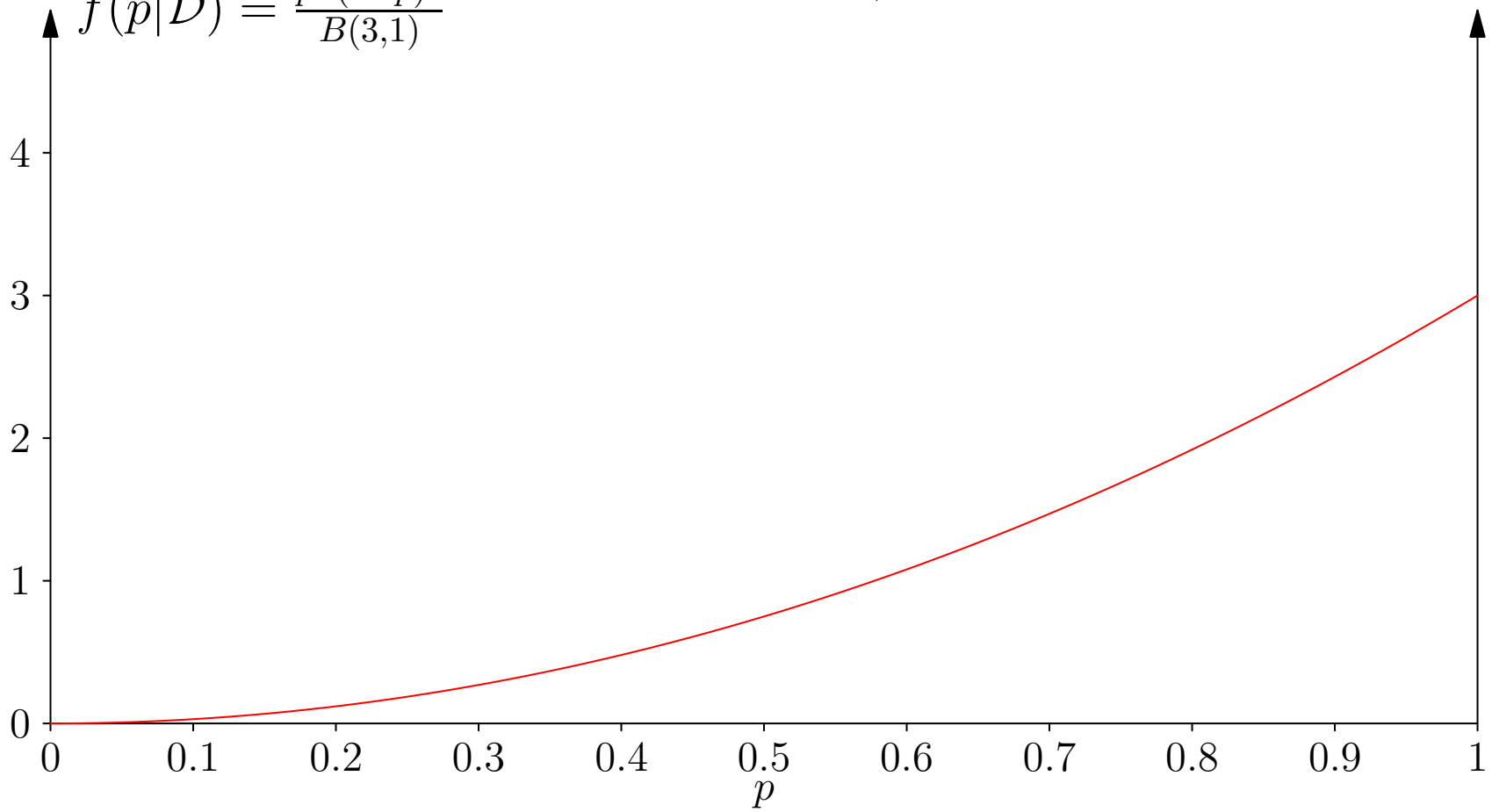


Example (p=0.7)

$\mathcal{D} = \{\text{H,H,}$

$f(p|\mathcal{D}) = \frac{p^2\,(1-p)^0}{B(3,1)}$

$a = 3,\ b = 1$

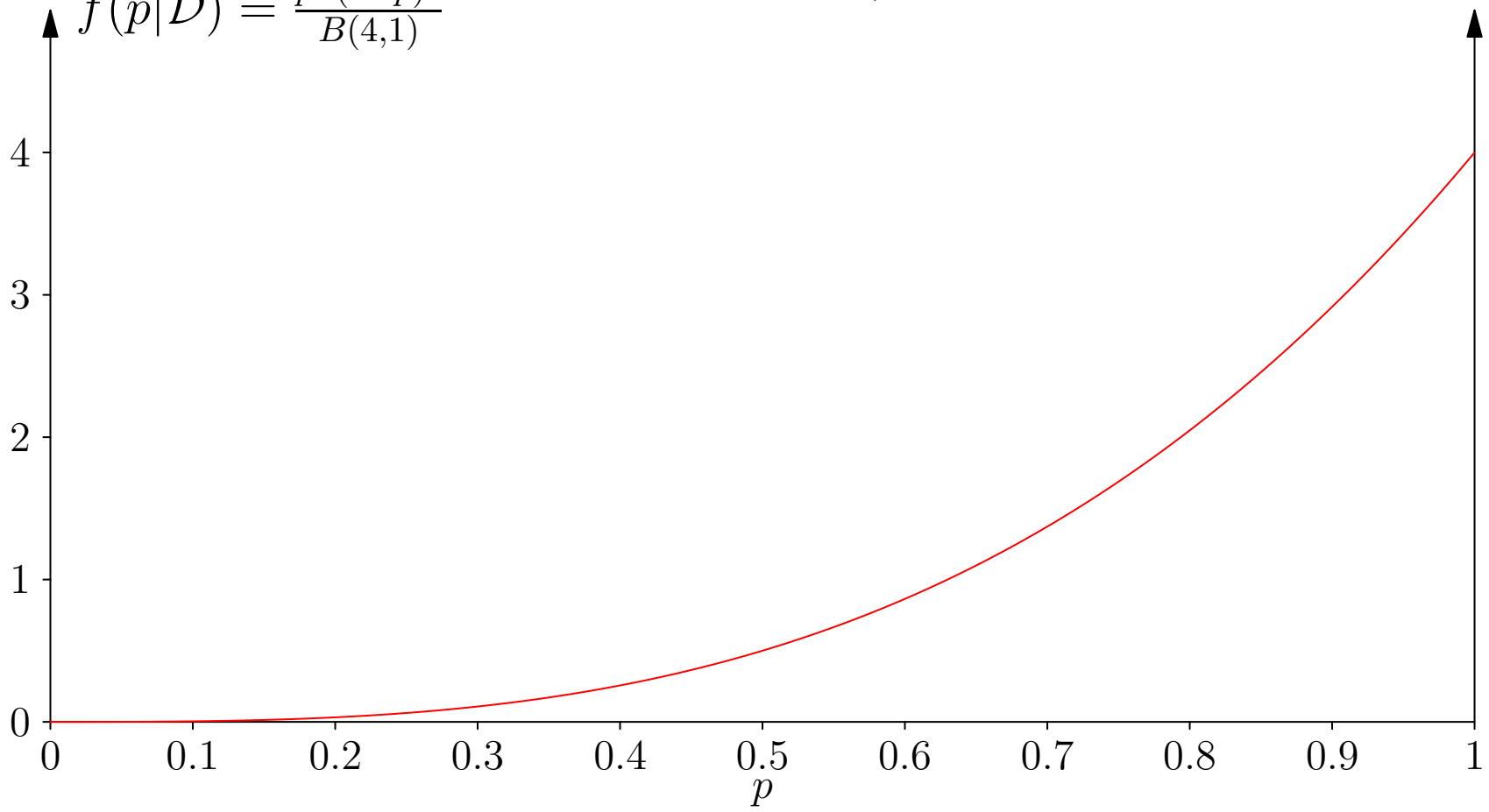


Example (p=0.7)

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$a = 4,\ b = 1$

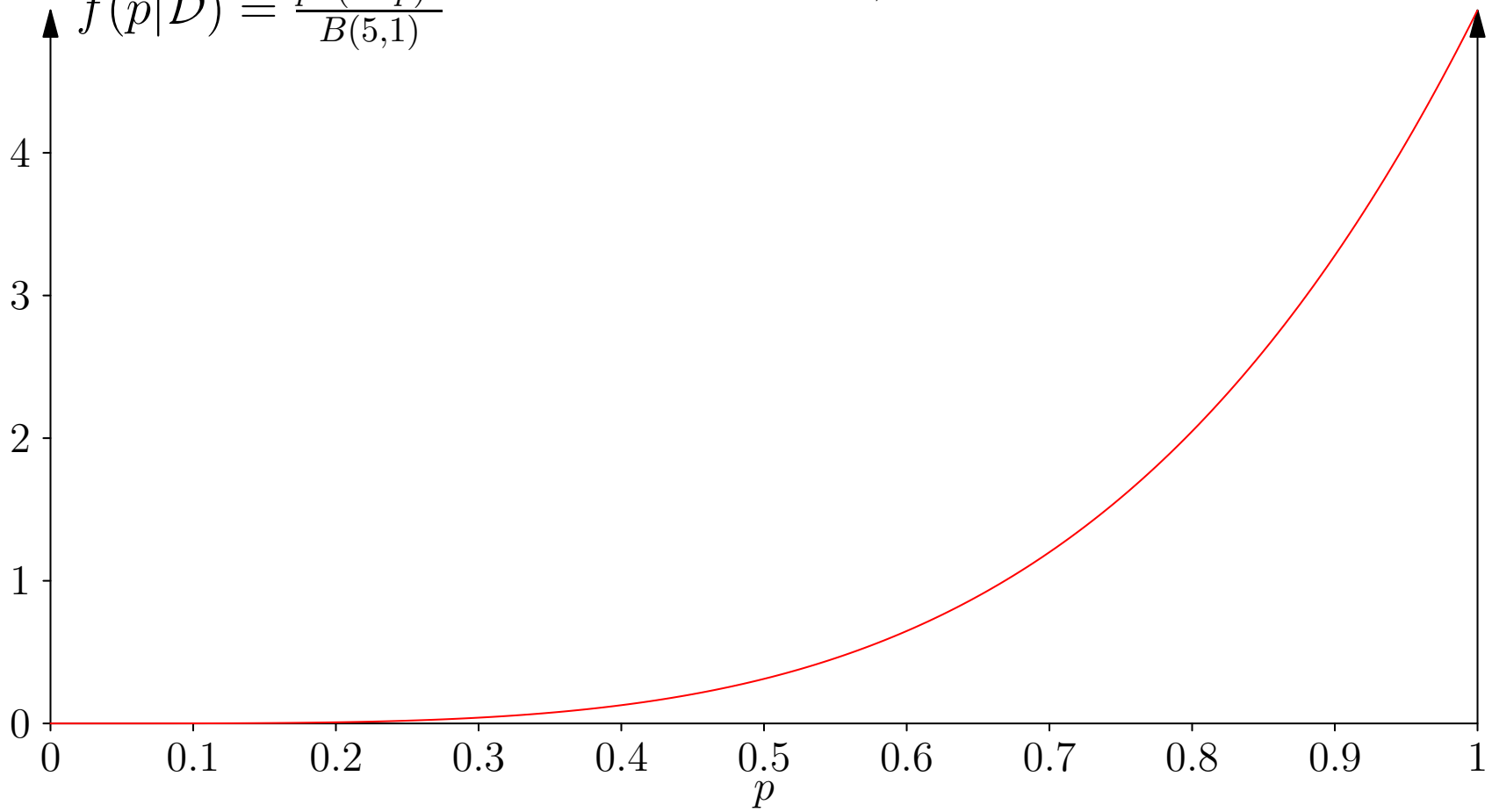


Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,}$

$f(p|\mathcal{D}) = \frac{p^4\,(1-p)^0}{B(5,1)}$

$a = 5,\ b = 1$

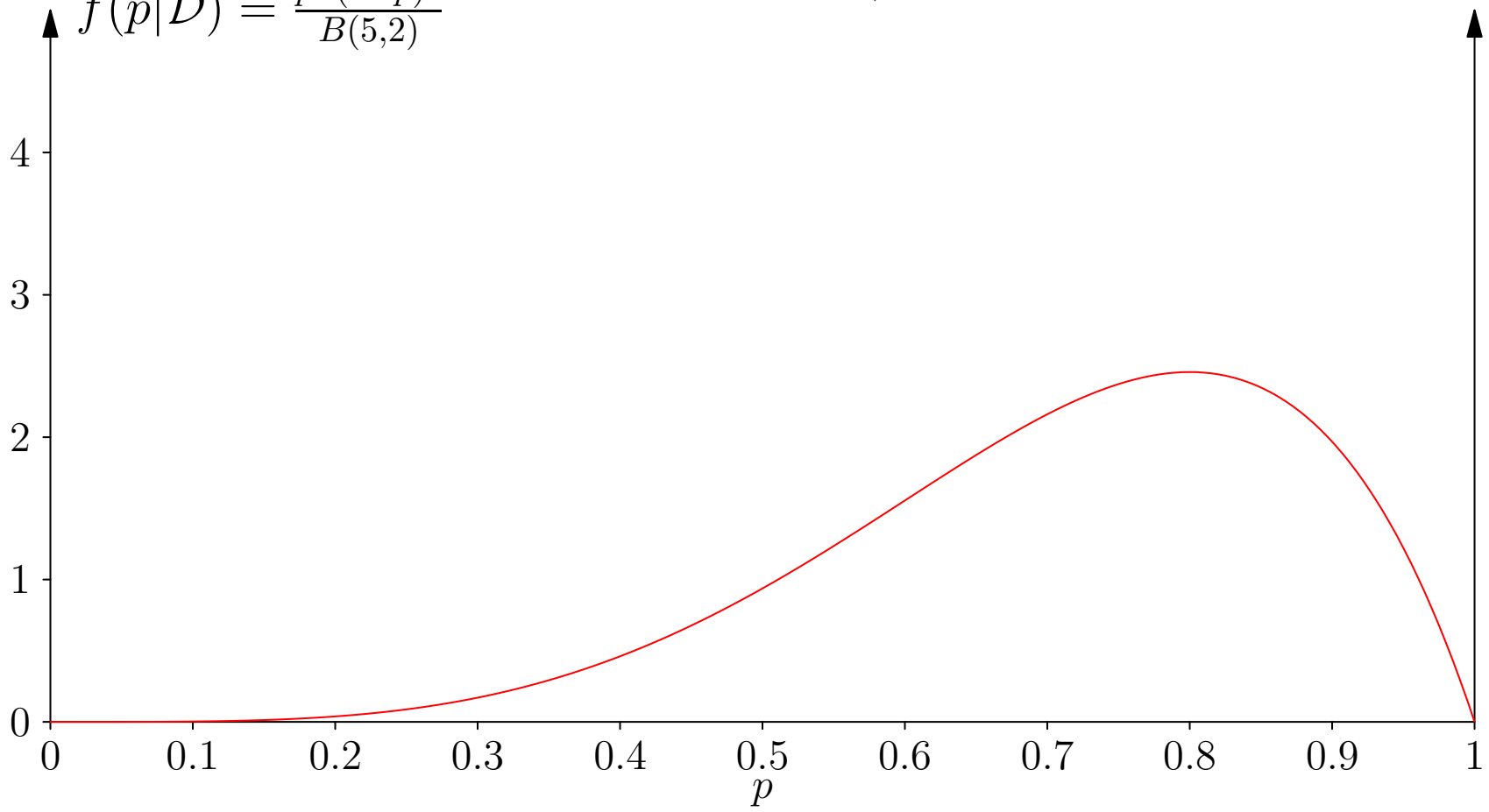


Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,}$

$f(p|\mathcal{D}) = \frac{p^4\,(1-p)^1}{B(5,2)}$

$a = 5,\ b = 2$

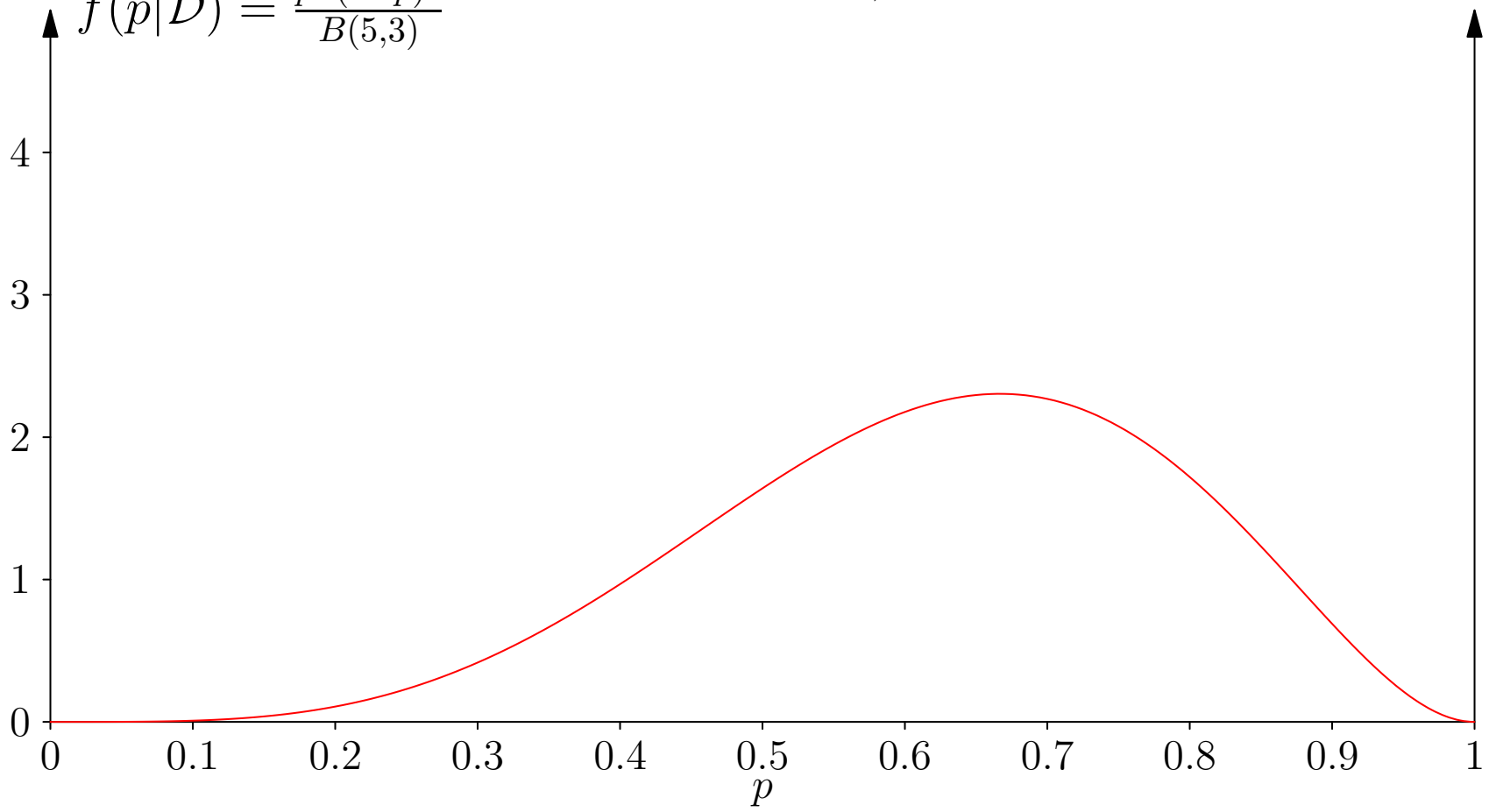


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$a = 5,\ b = 3$

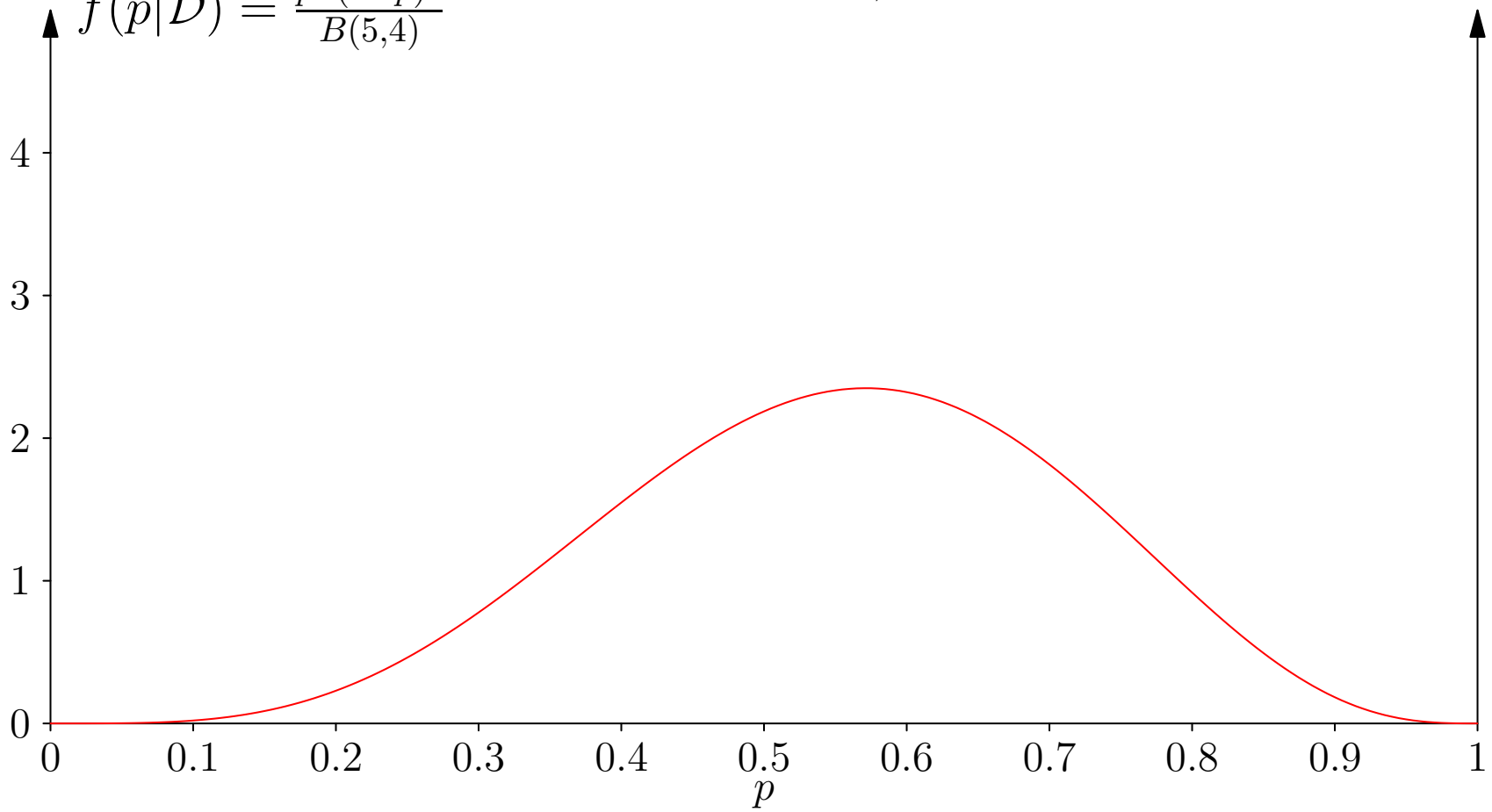


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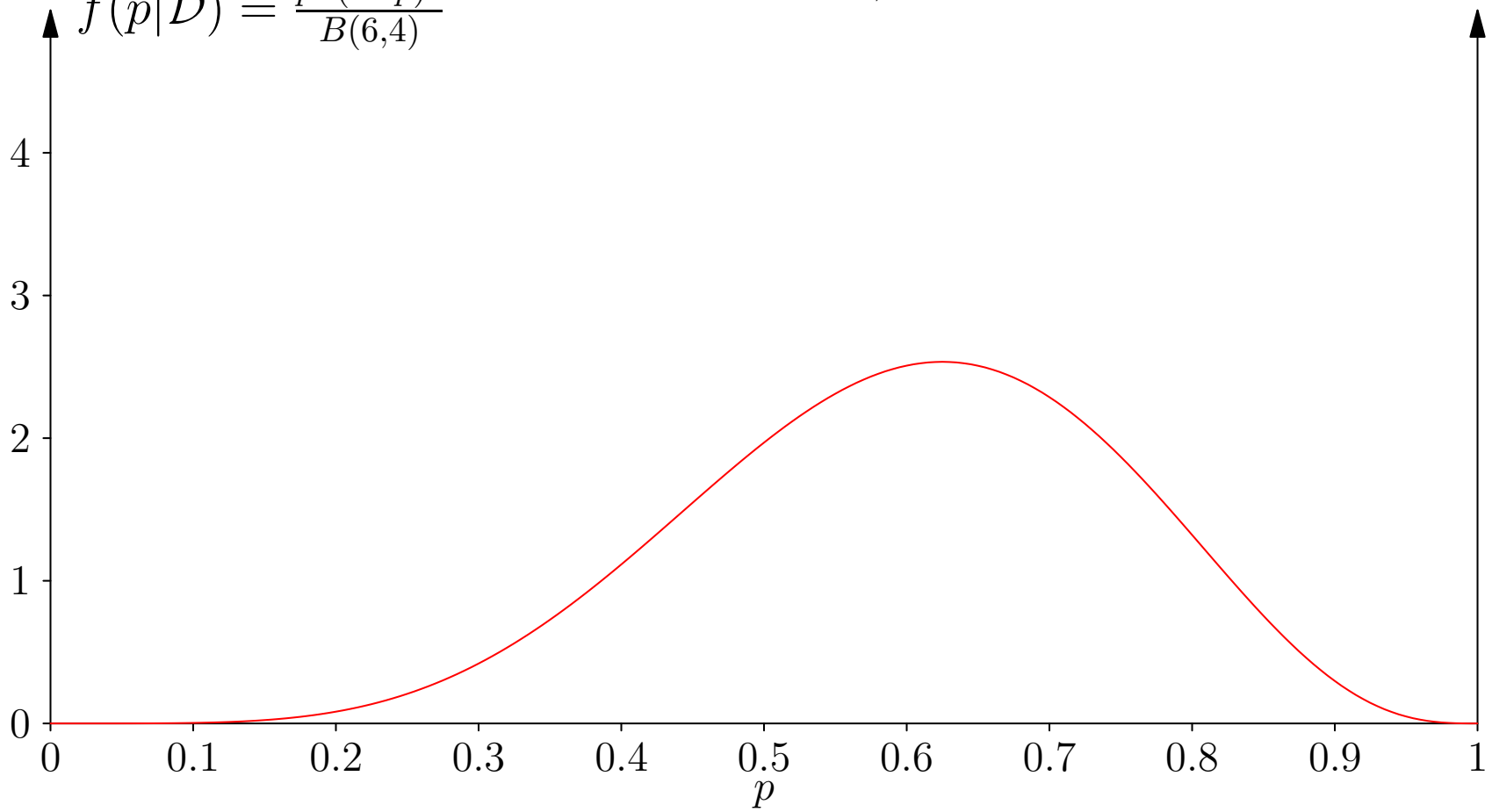


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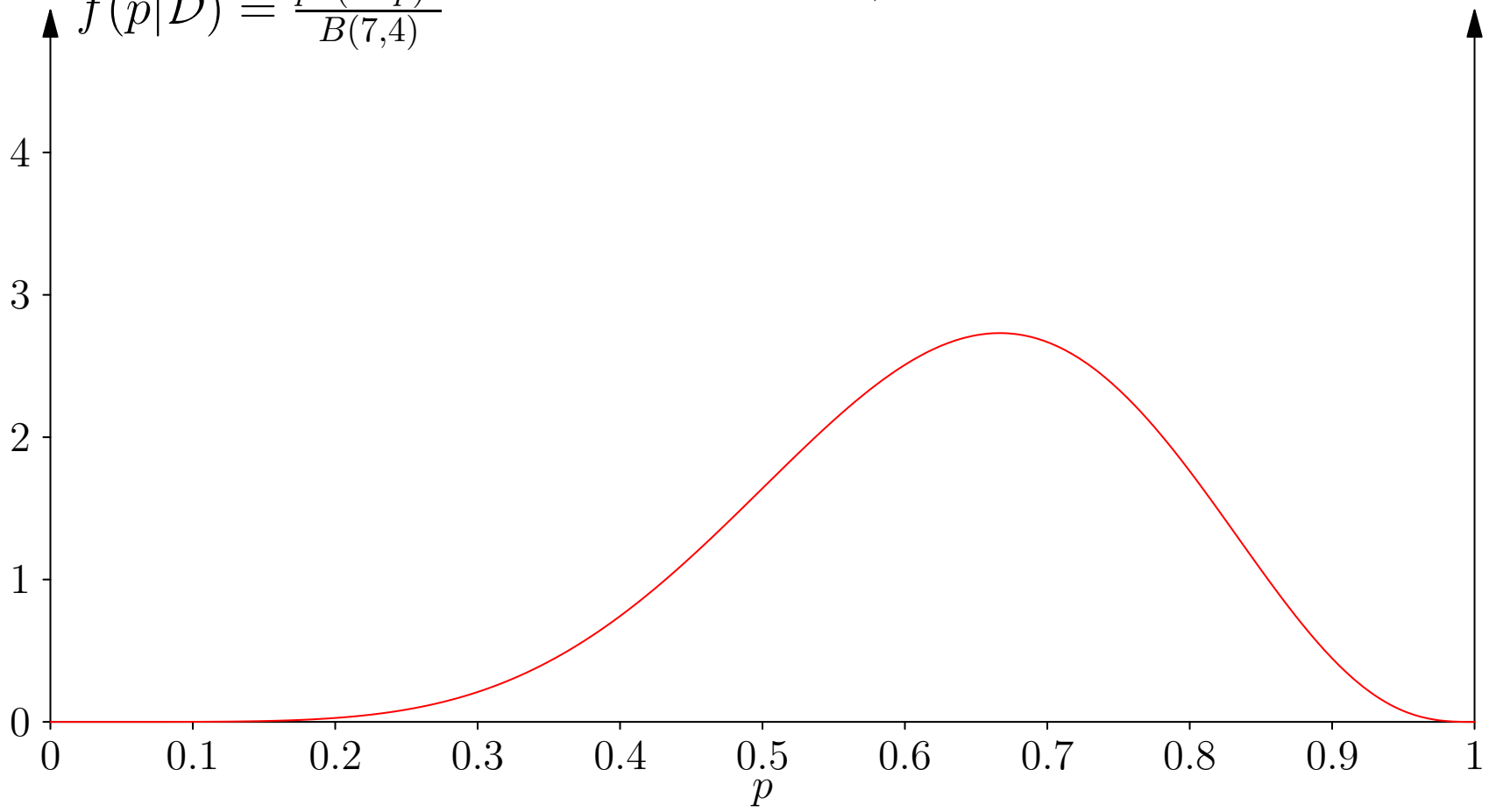
$a = 6,\ b = 4$



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$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,}$

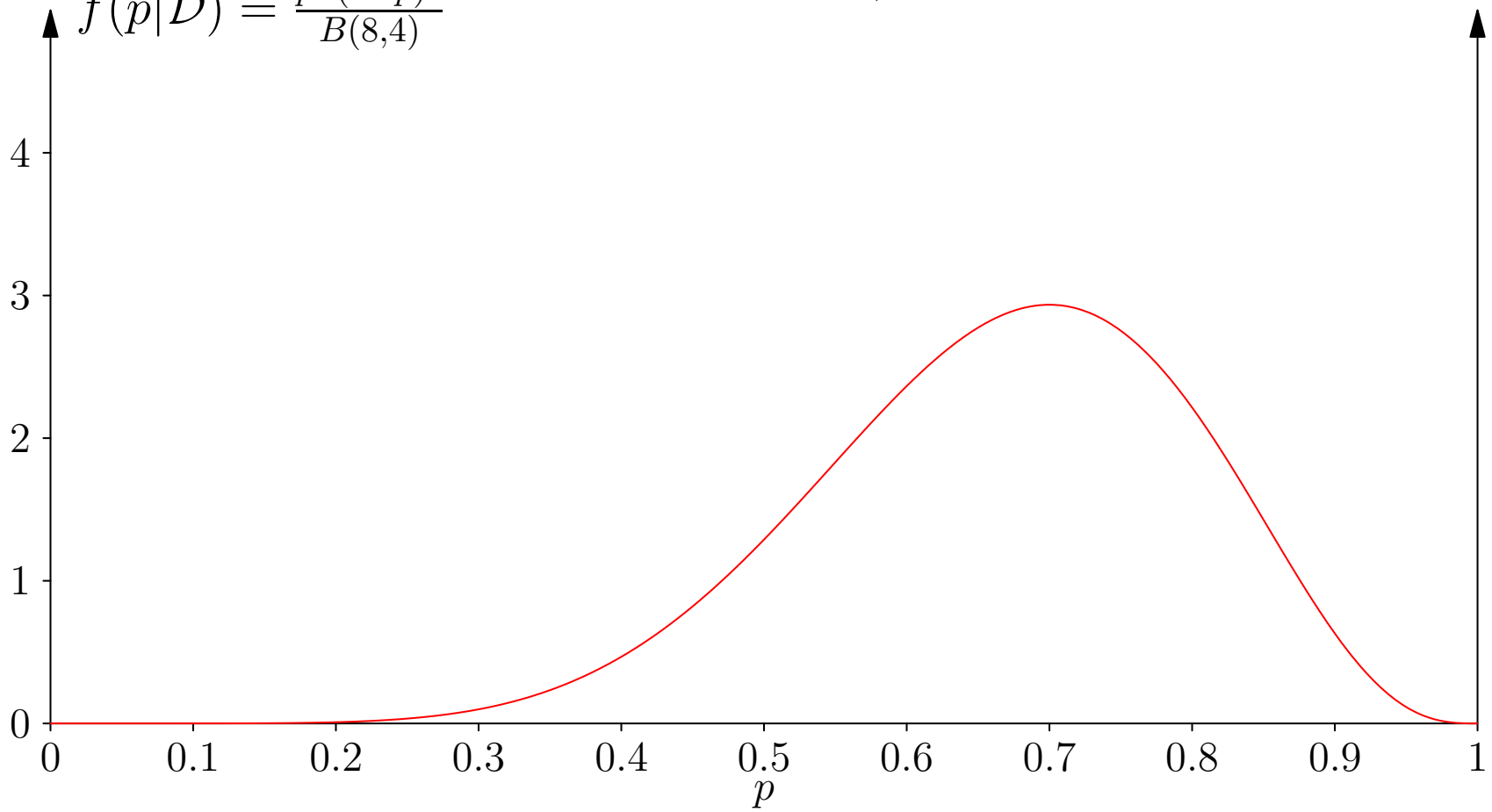
$f(p|\mathcal{D}) = \frac{p^6\,(1-p)^3}{B(7,4)}$ $\qquad a = 7,\ b = 4$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,}$

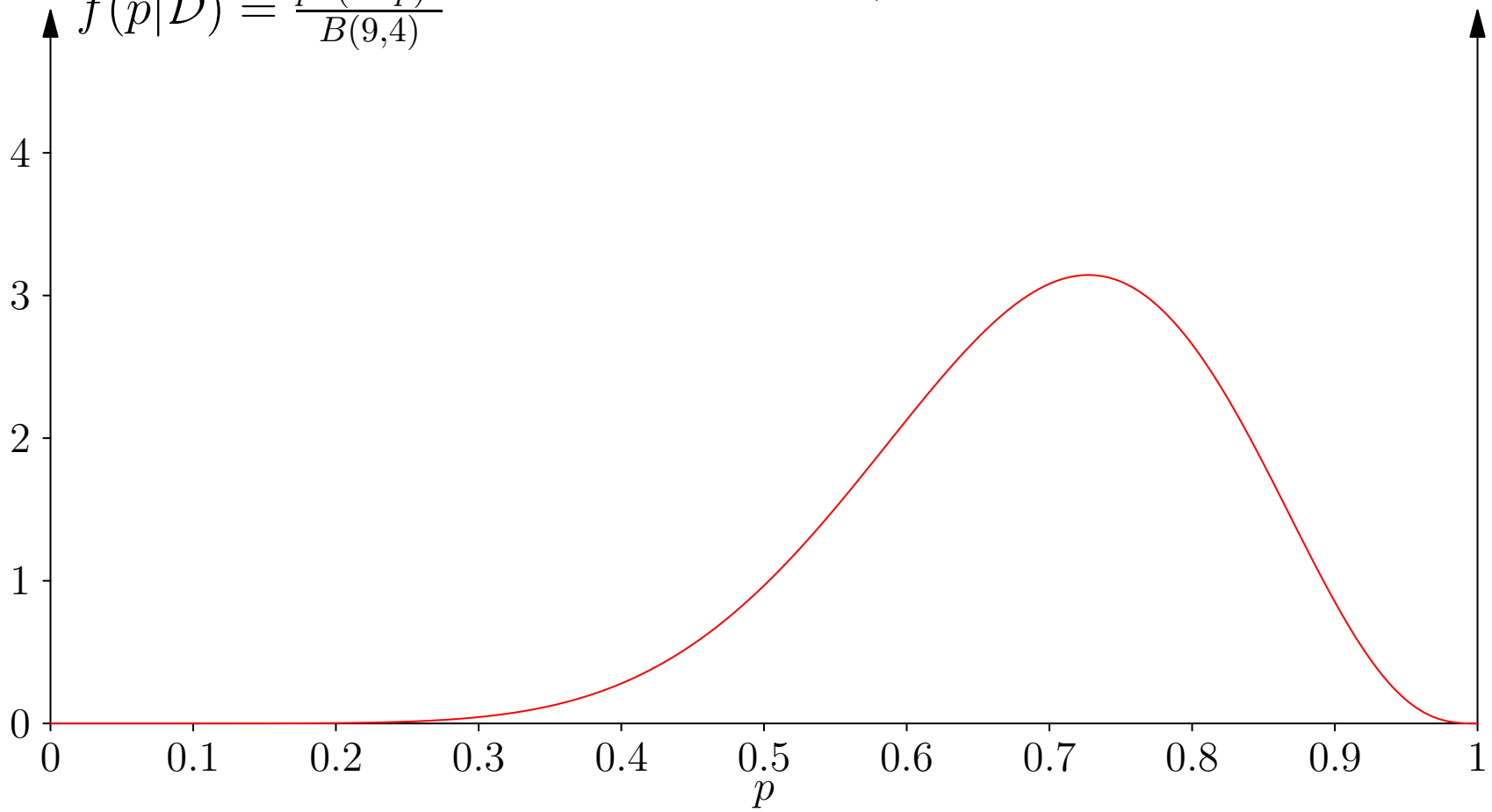
$f(p|\mathcal{D}) = \frac{p^7\,(1-p)^3}{B(8,4)}$ $\qquad a = 8,\ b = 4$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,}$

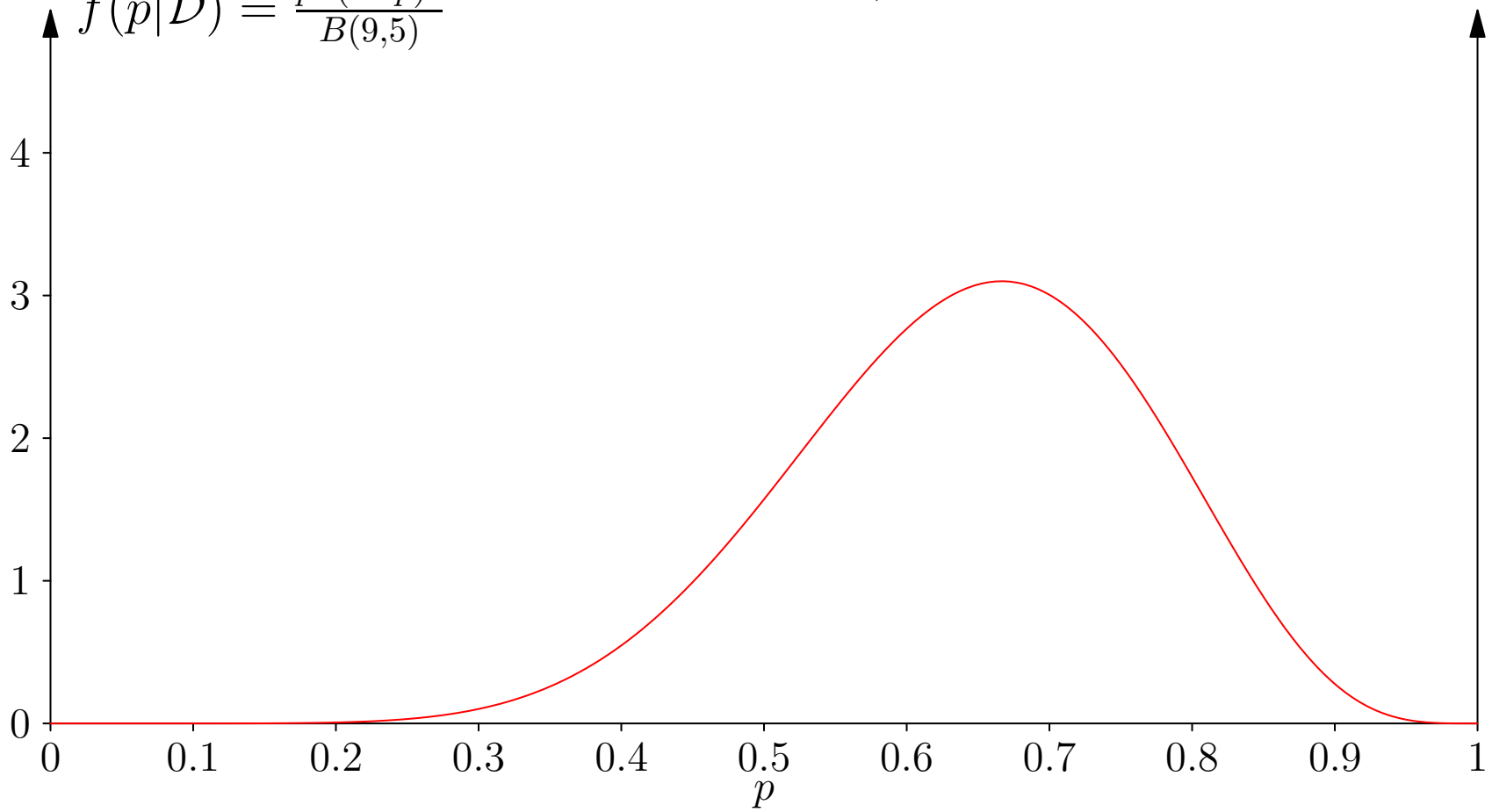
$f(p|\mathcal{D}) = \frac{p^8\,(1-p)^3}{B(9,4)}$ $\quad a = 9,\ b = 4$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,}$

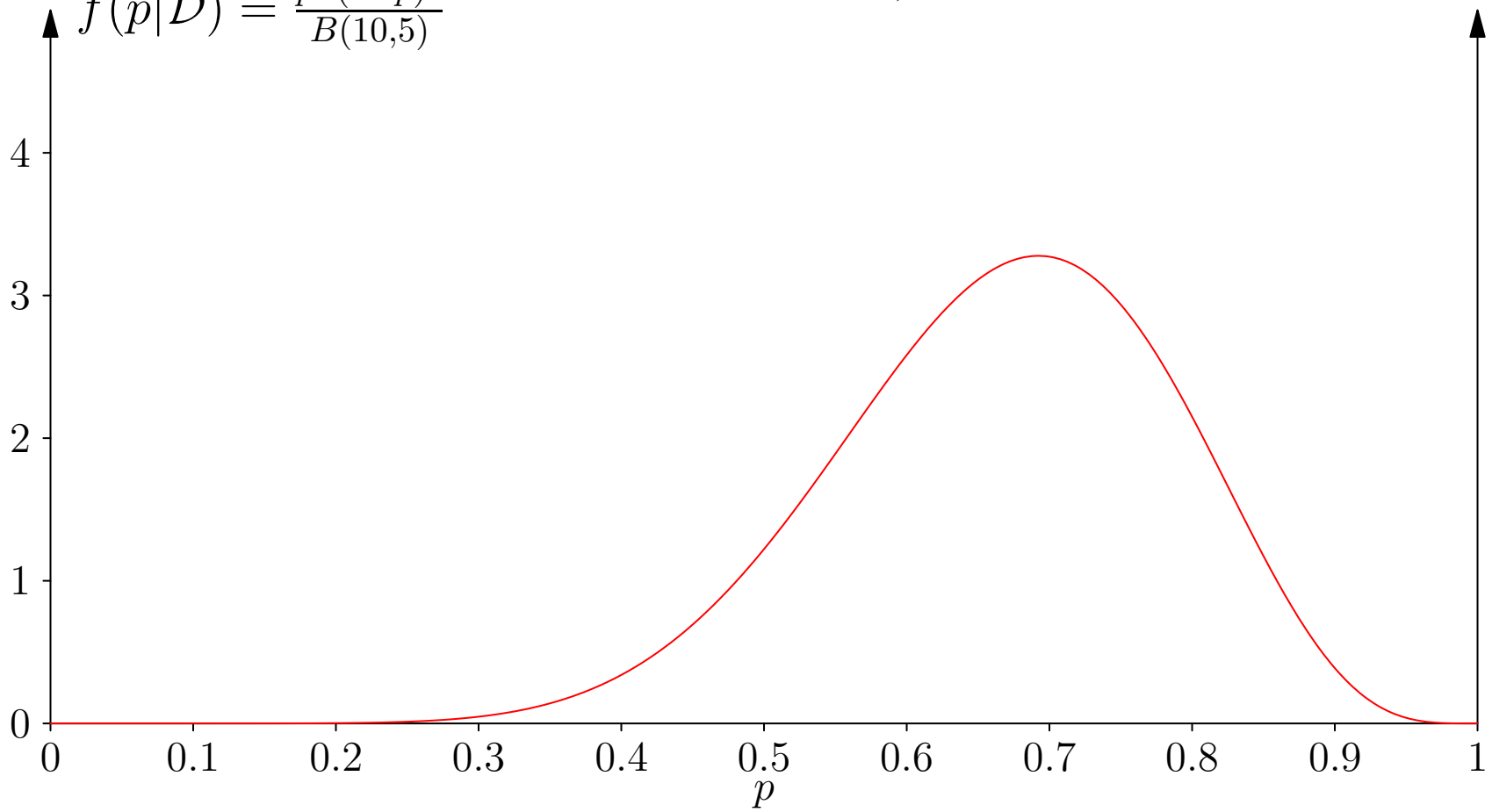
$f(p|\mathcal{D}) = \frac{p^8\,(1-p)^4}{B(9,5)}$ $\qquad a = 9,\ b = 5$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,}$

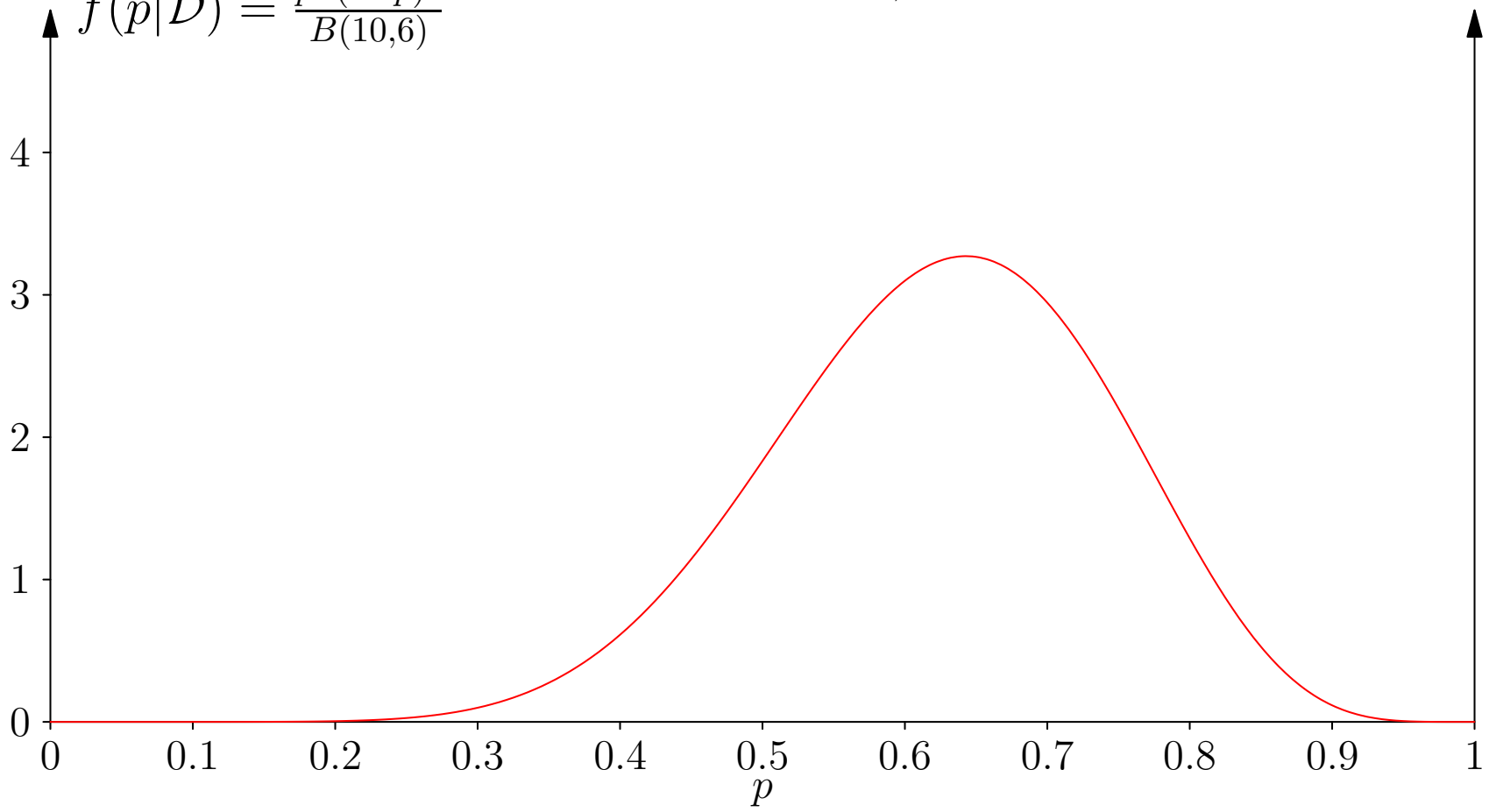
$f(p|\mathcal{D}) = \frac{p^9\,(1-p)^4}{B(10,5)}$ $\qquad a = 10,\ b = 5$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,}$

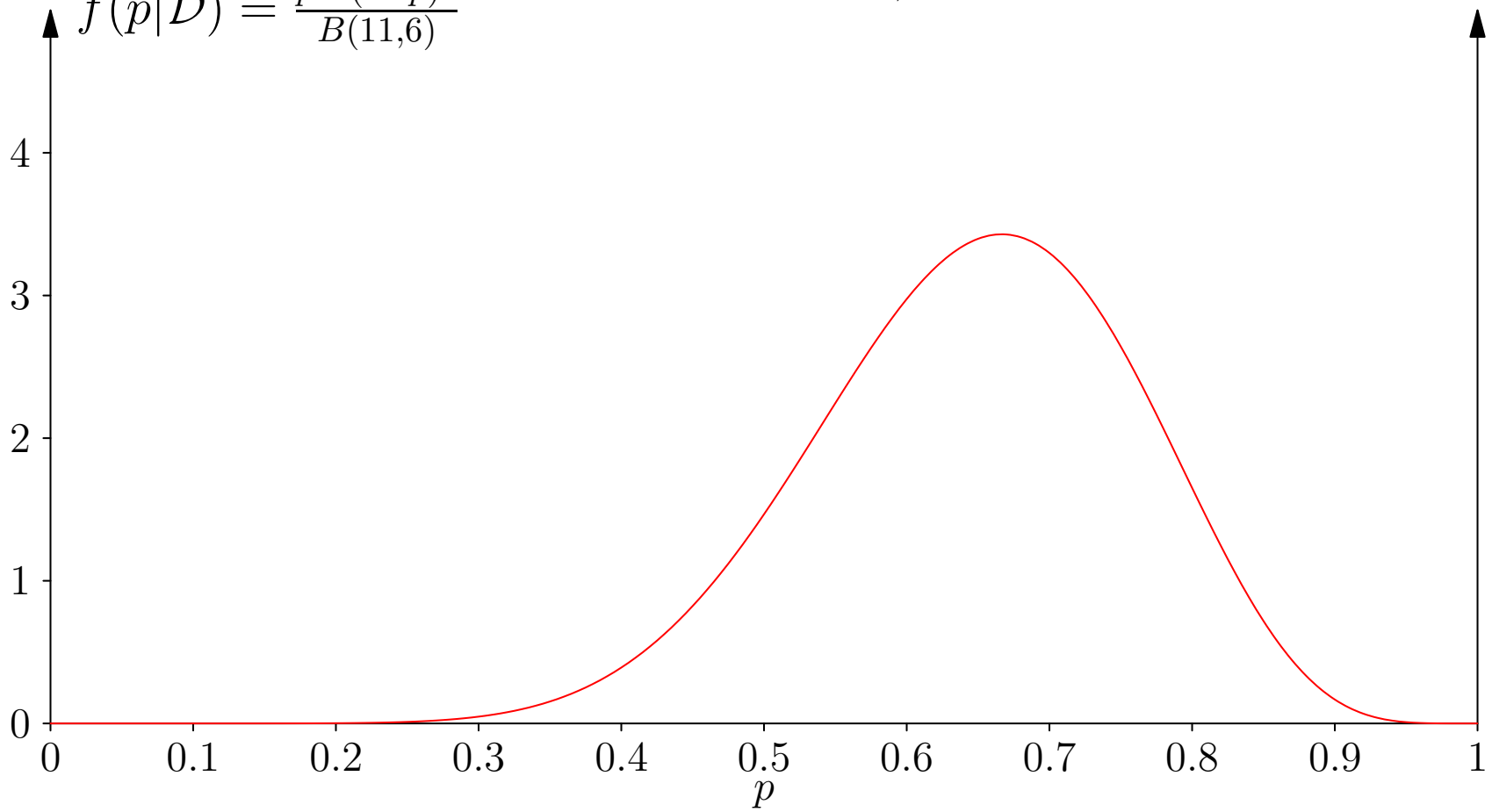
$f(p|\mathcal{D}) = \frac{p^9\,(1-p)^5}{B(10{,}6)}$ $\qquad a = 10,\ b = 6$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,}$

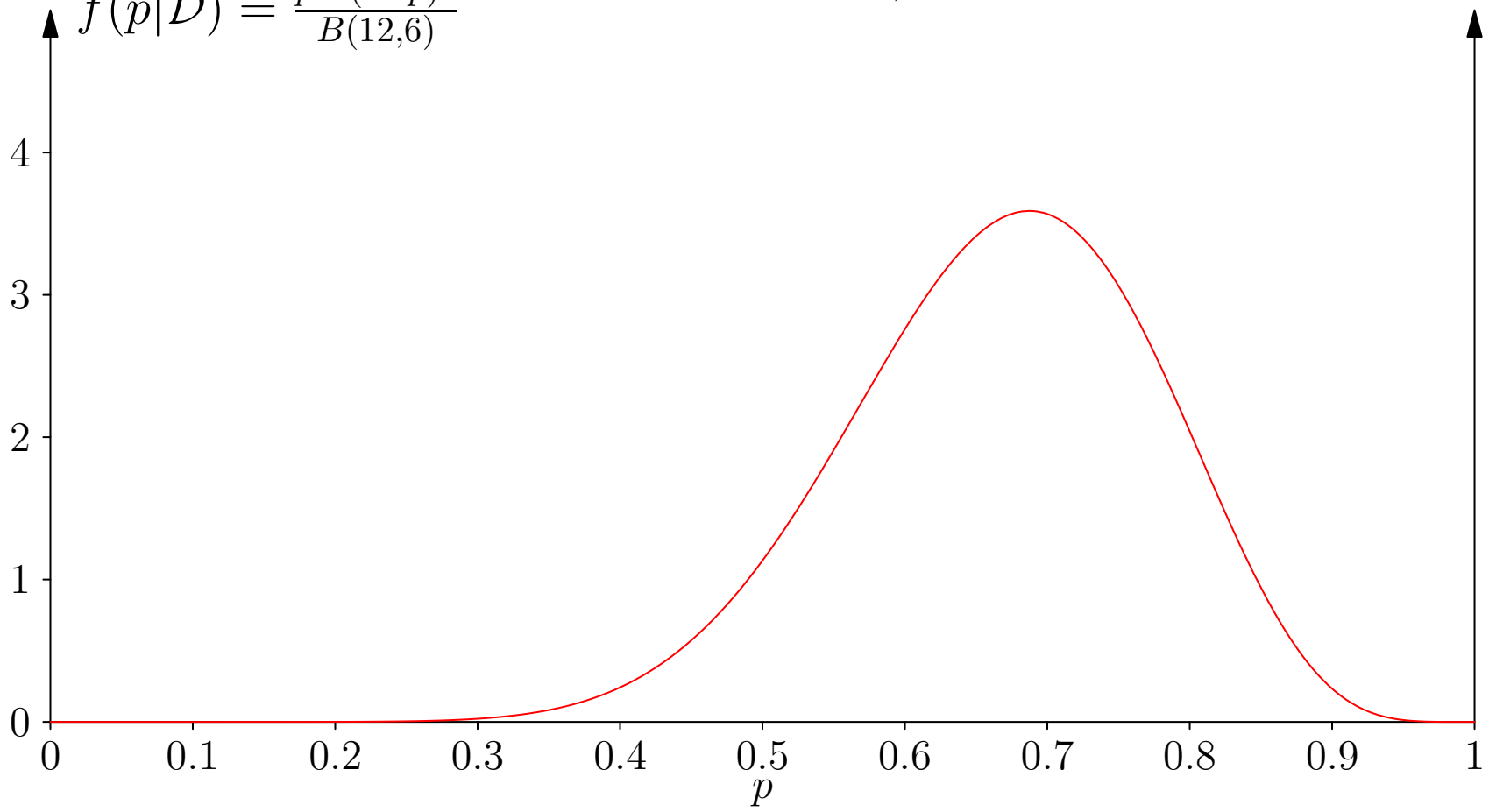
$f(p|\mathcal{D}) = \frac{p^{10}\,(1-p)^5}{B(11,6)}$ $\qquad a = 11,\ b = 6$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,}$

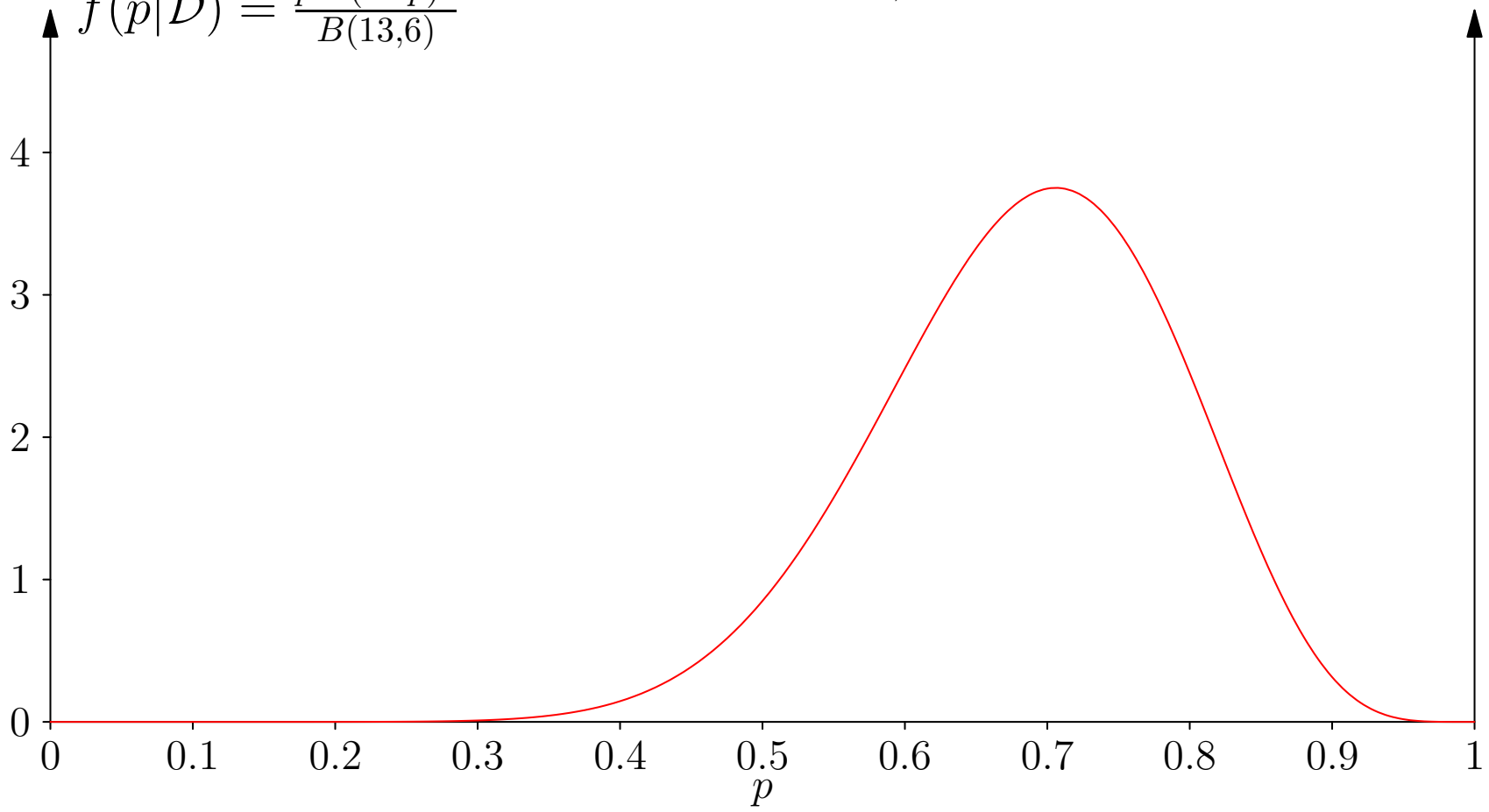
$f(p|\mathcal{D}) = \frac{p^{11}\,(1-p)^5}{B(12,6)}$ $\qquad a = 12,\ b = 6$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,}$

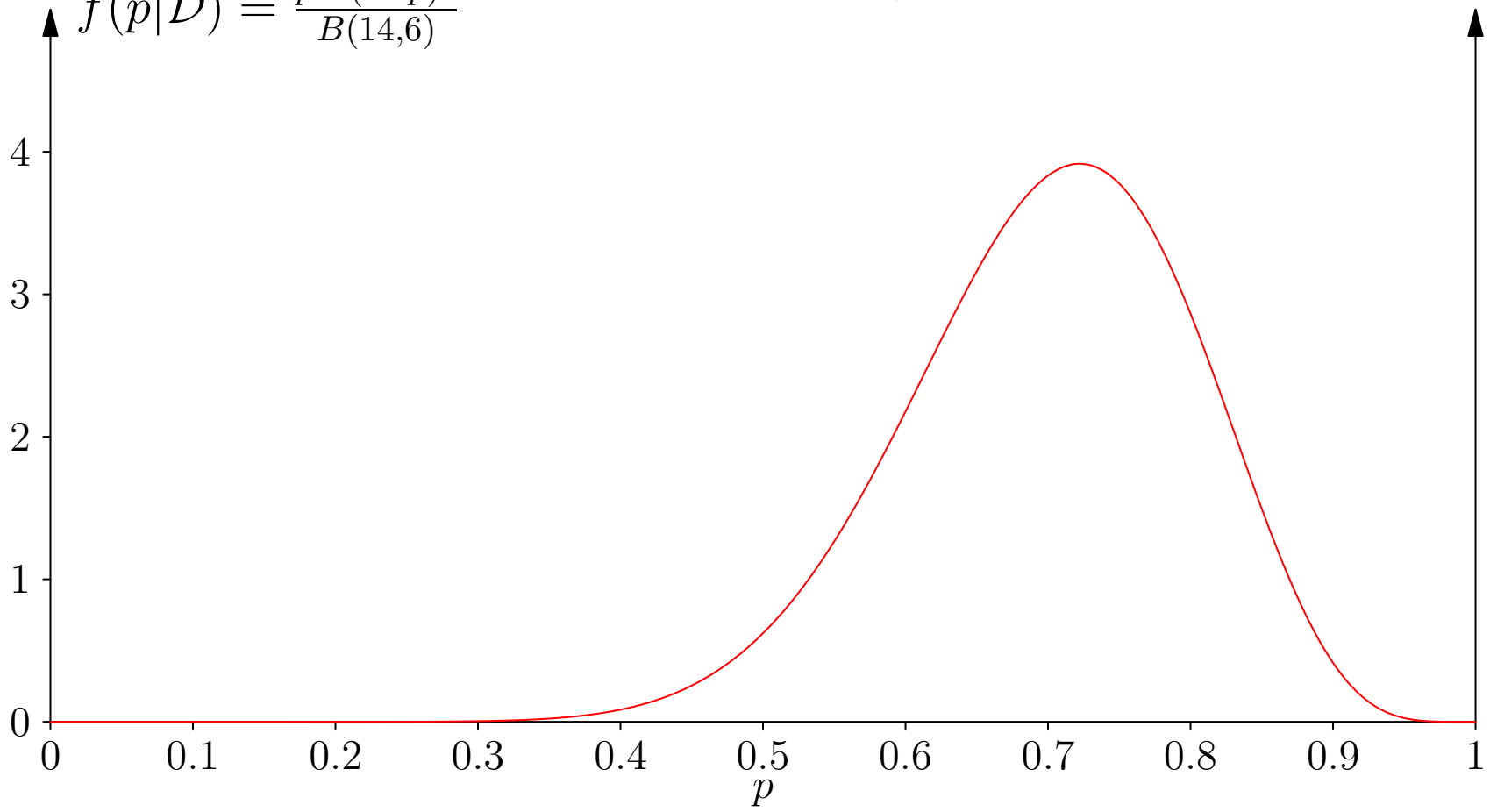
$f(p|\mathcal{D}) = \frac{p^{12}\,(1-p)^5}{B(13,6)}$ $\qquad a = 13,\ b = 6$



Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,}$$

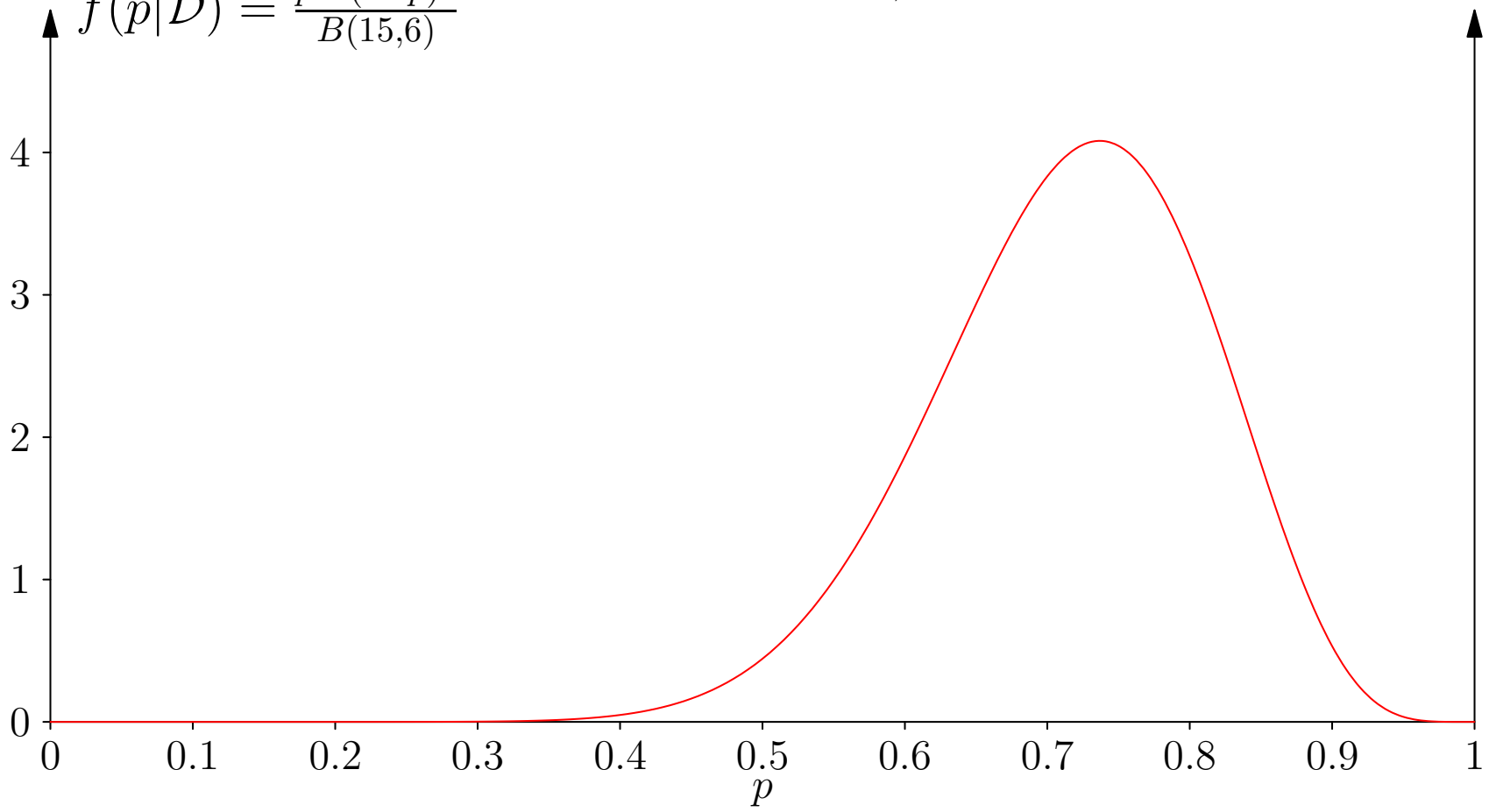
$$f(p|\mathcal{D}) = \frac{p^{13}\,(1-p)^5}{B(14,6)} \qquad a = 14,\ b = 6$$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,}$

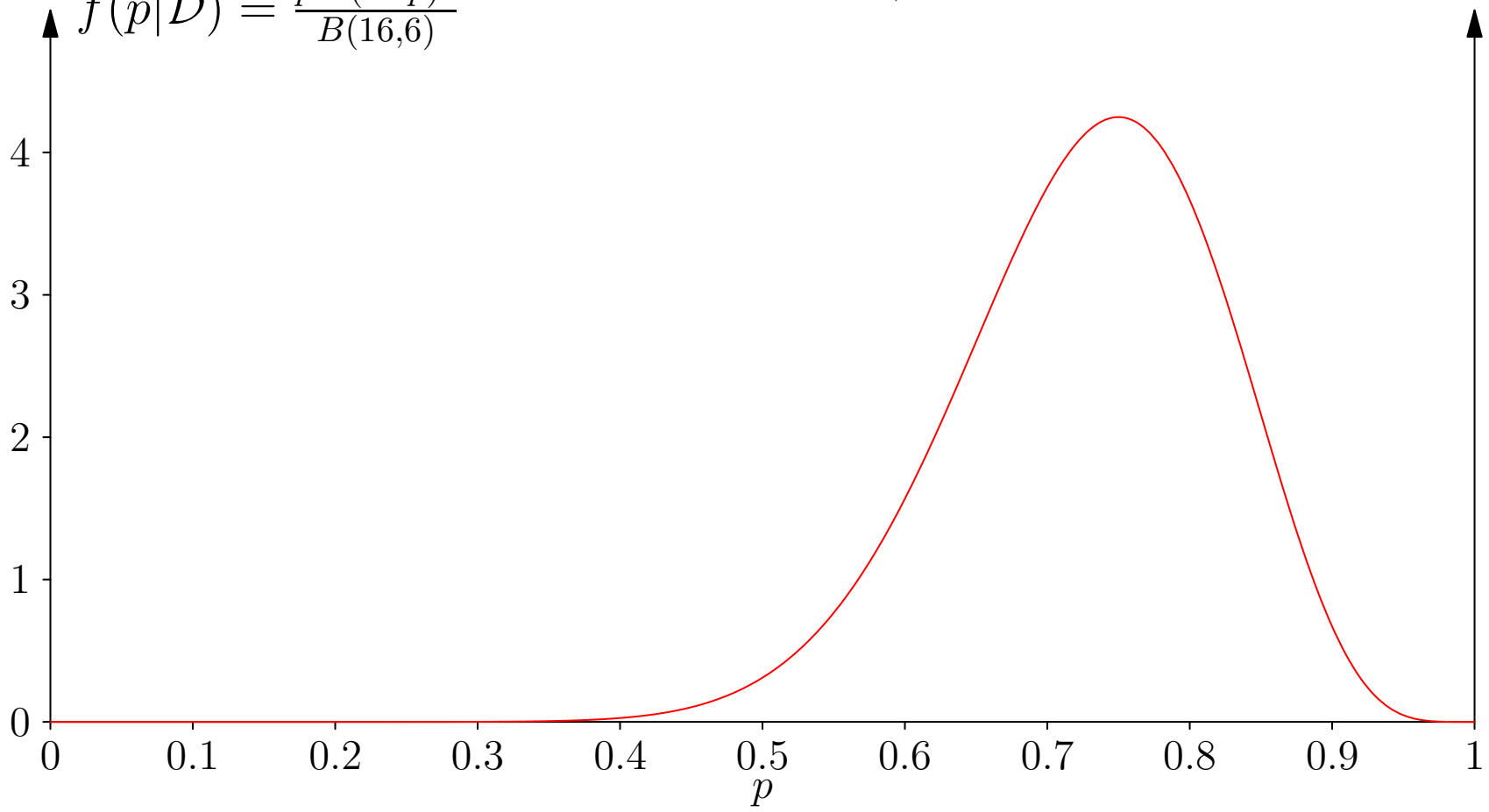
$f(p|\mathcal{D}) = \frac{p^{14}\,(1-p)^5}{B(15,6)}$ $\qquad a = 15,\ b = 6$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,}$

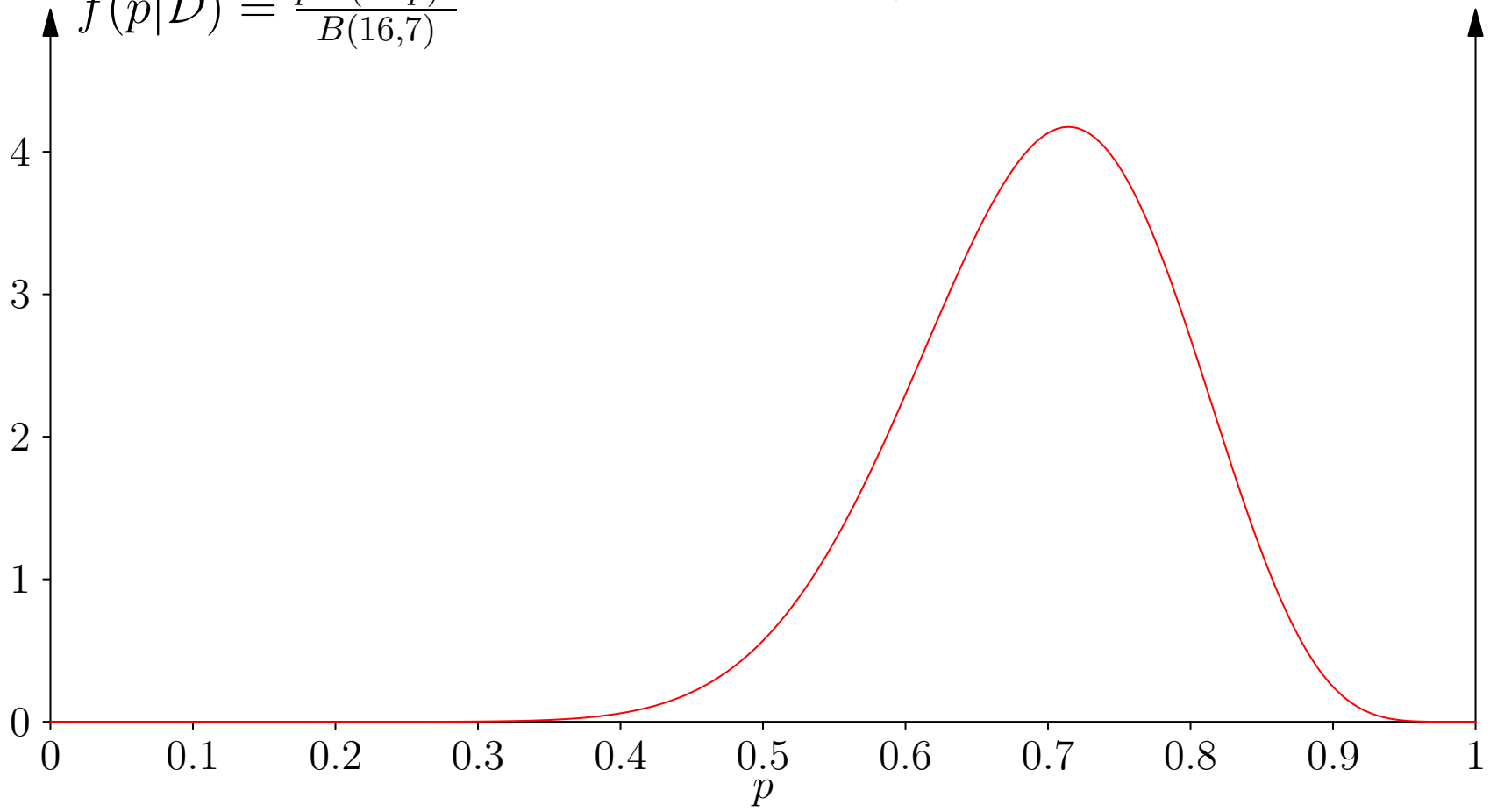
$f(p|\mathcal{D}) = \frac{p^{15}\,(1-p)^5}{B(16,6)}$ $\qquad a = 16,\ b = 6$



Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,}$$

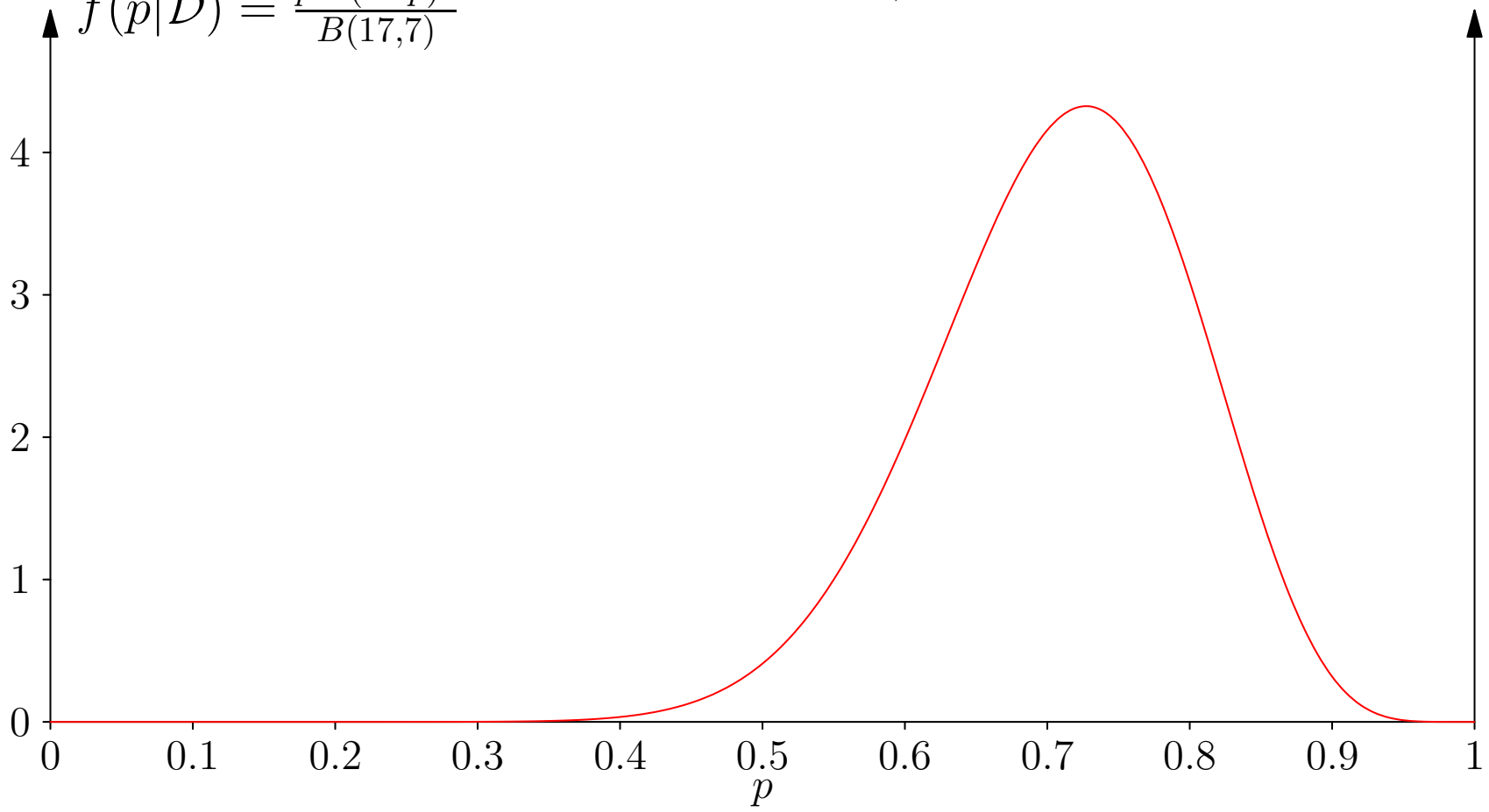
$$f(p|\mathcal{D}) = \frac{p^{15}\,(1-p)^6}{B(16,7)} \qquad a = 16,\ b = 7$$



Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,}$$

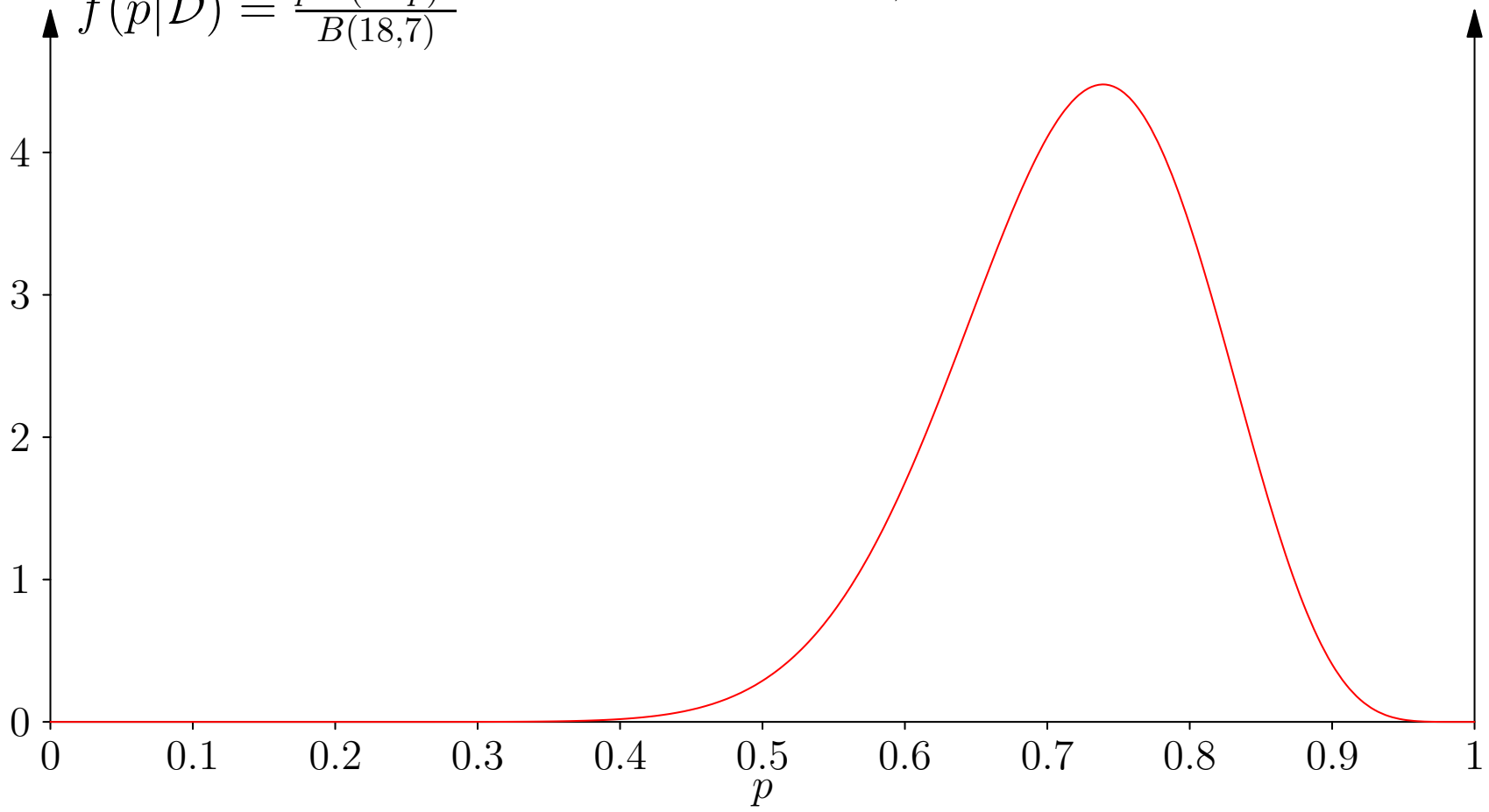
$$f(p|\mathcal{D}) = \frac{p^{16}\,(1-p)^6}{B(17,7)} \qquad a = 17,\ b = 7$$



Example (p=0.7)

$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,H,}$

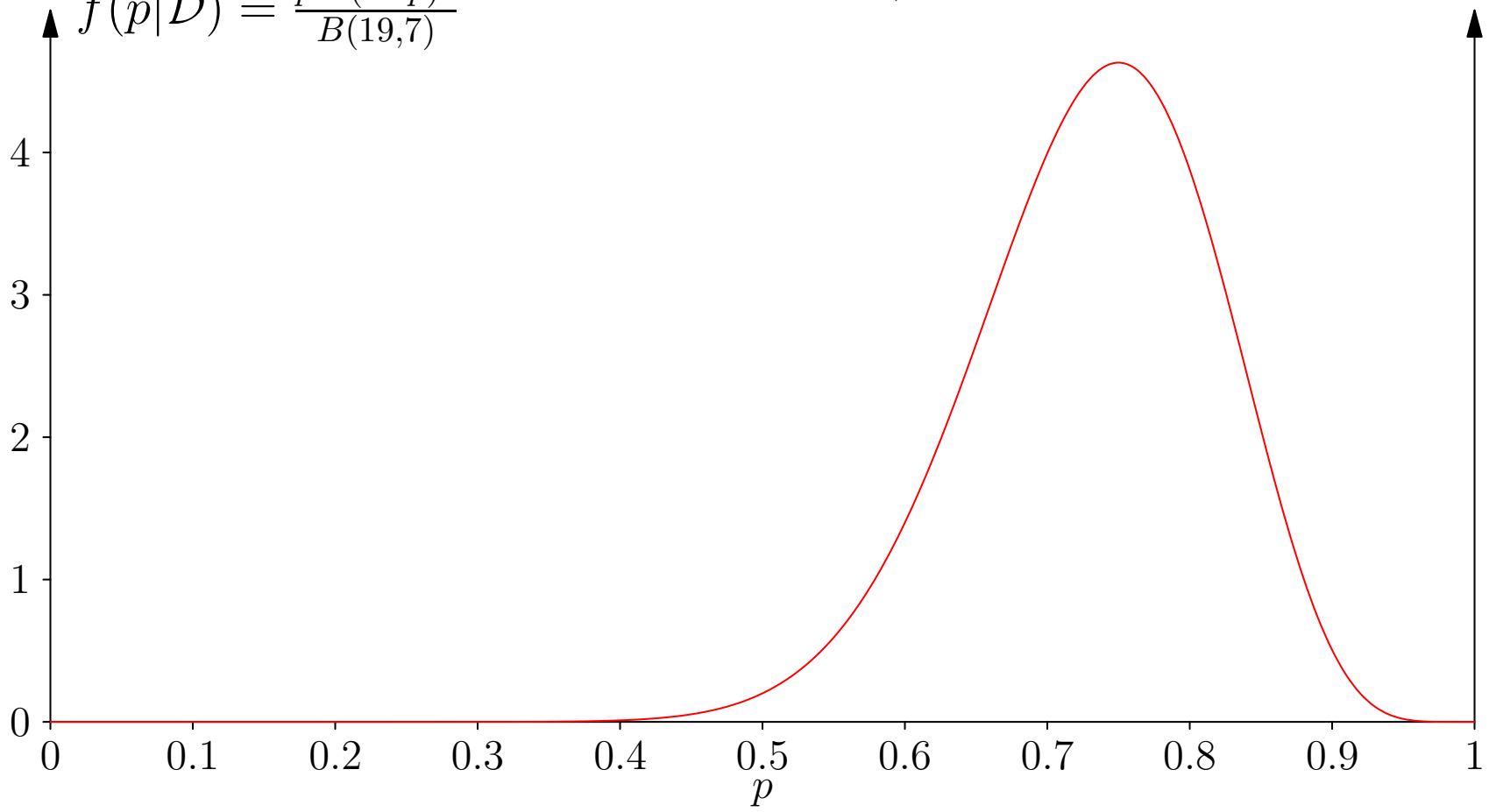
$f(p|\mathcal{D}) = \frac{p^{17}\,(1-p)^6}{B(18,7)}$ $\quad a = 18,\ b = 7$



Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,H,H,}$$

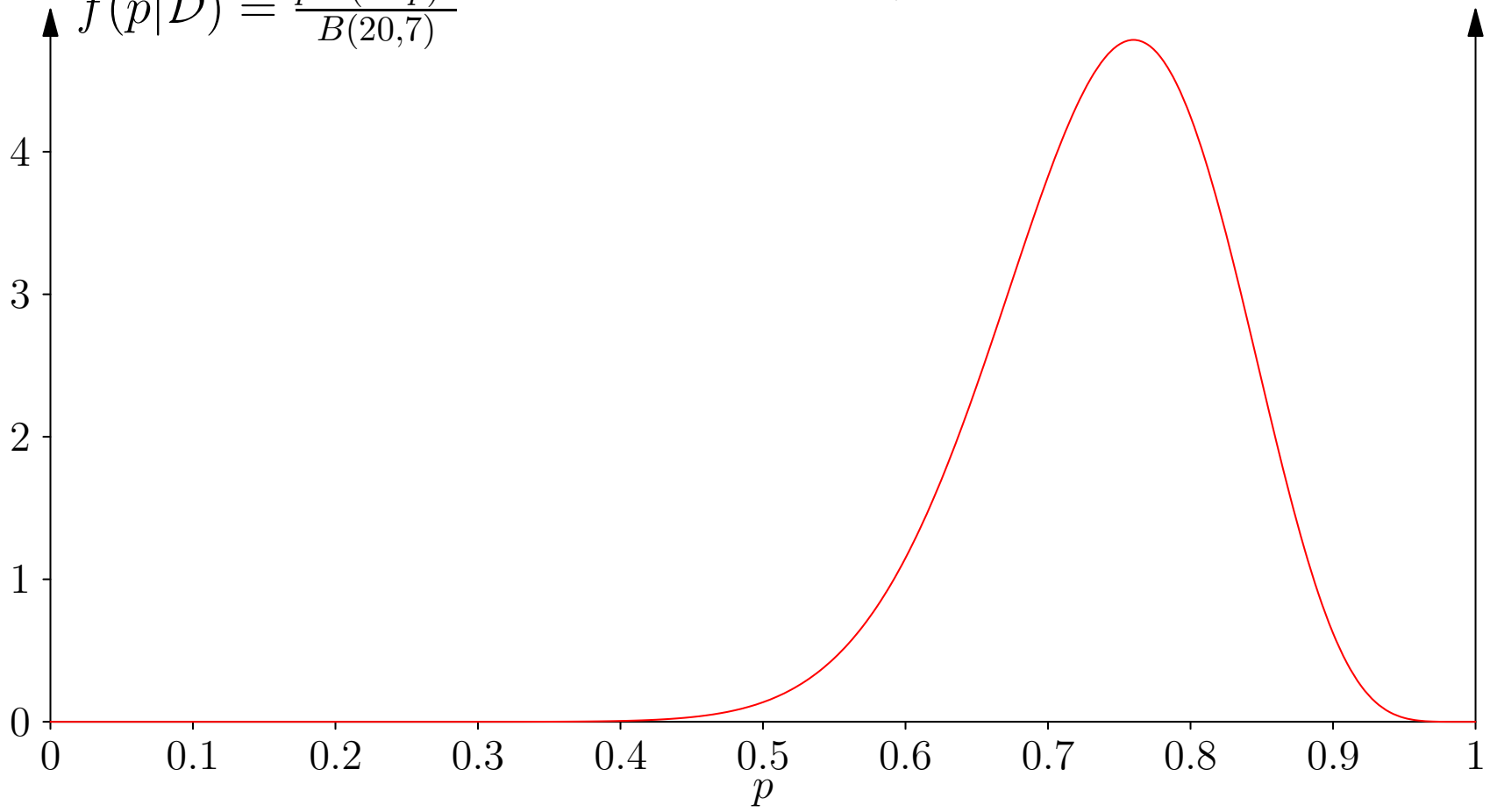
$$f(p|\mathcal{D}) = \frac{p^{18}\,(1-p)^{6}}{B(19,7)} \qquad a = 19,\ b = 7$$



Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,H,H,H,}$$

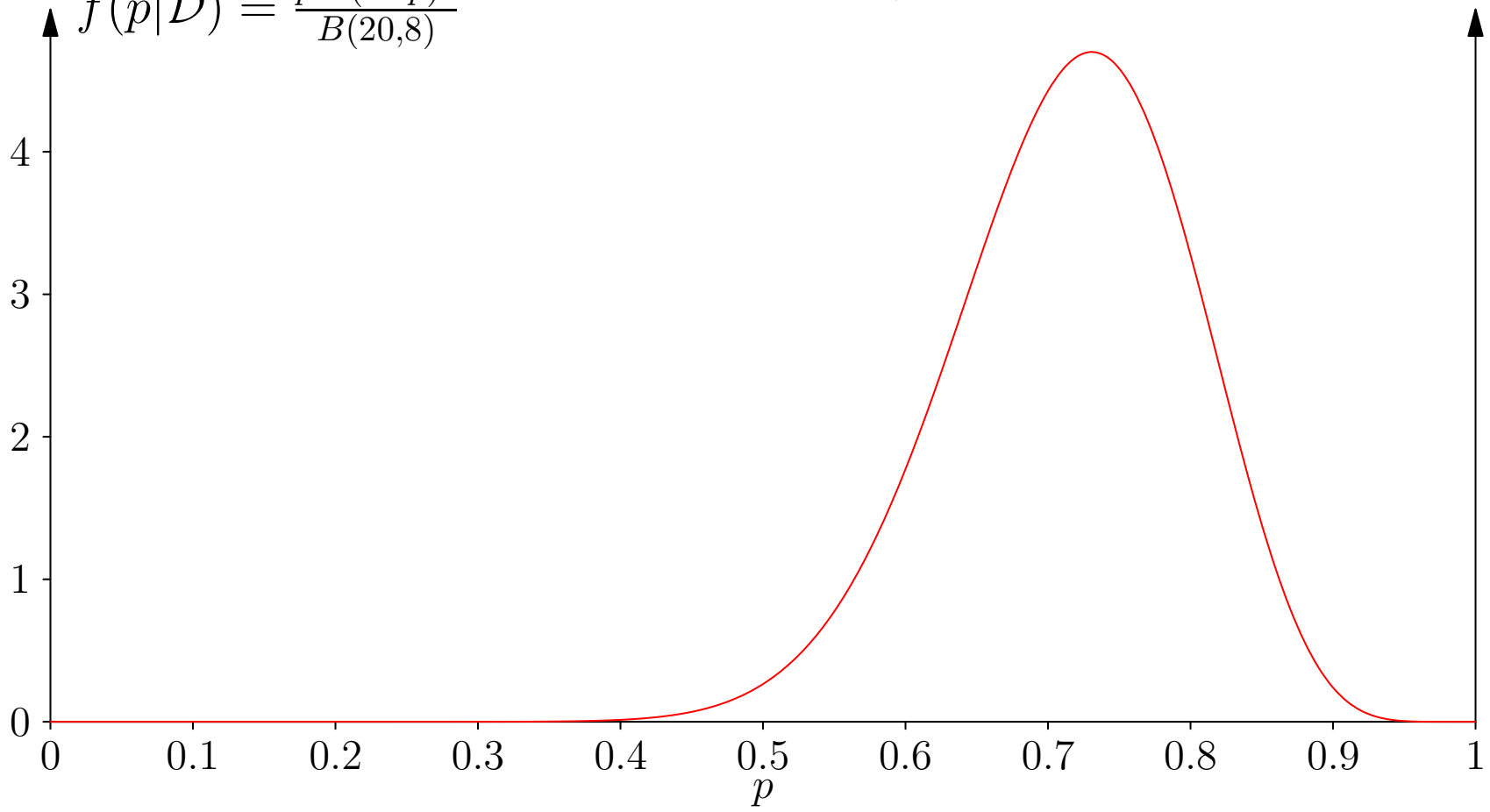
$$f(p|\mathcal{D}) = \frac{p^{19}\,(1-p)^{6}}{B(20,7)} \qquad a = 20,\ b = 7$$



Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,H,H,H,T,}$$

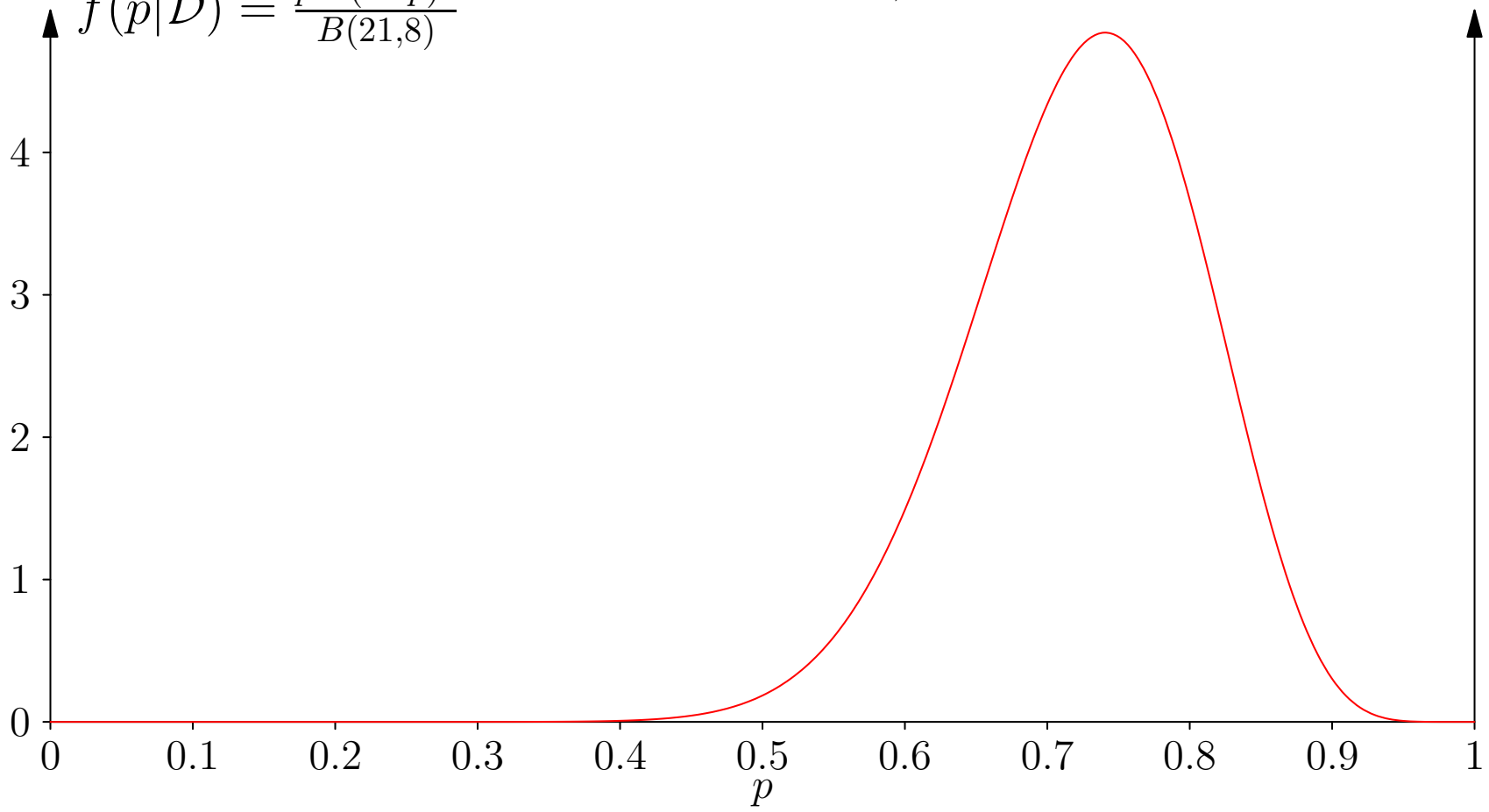
$$f(p|\mathcal{D}) = \frac{p^{19}\,(1-p)^7}{B(20,8)} \qquad a = 20,\ b = 8$$



Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,H,H,H,T,H,}$$

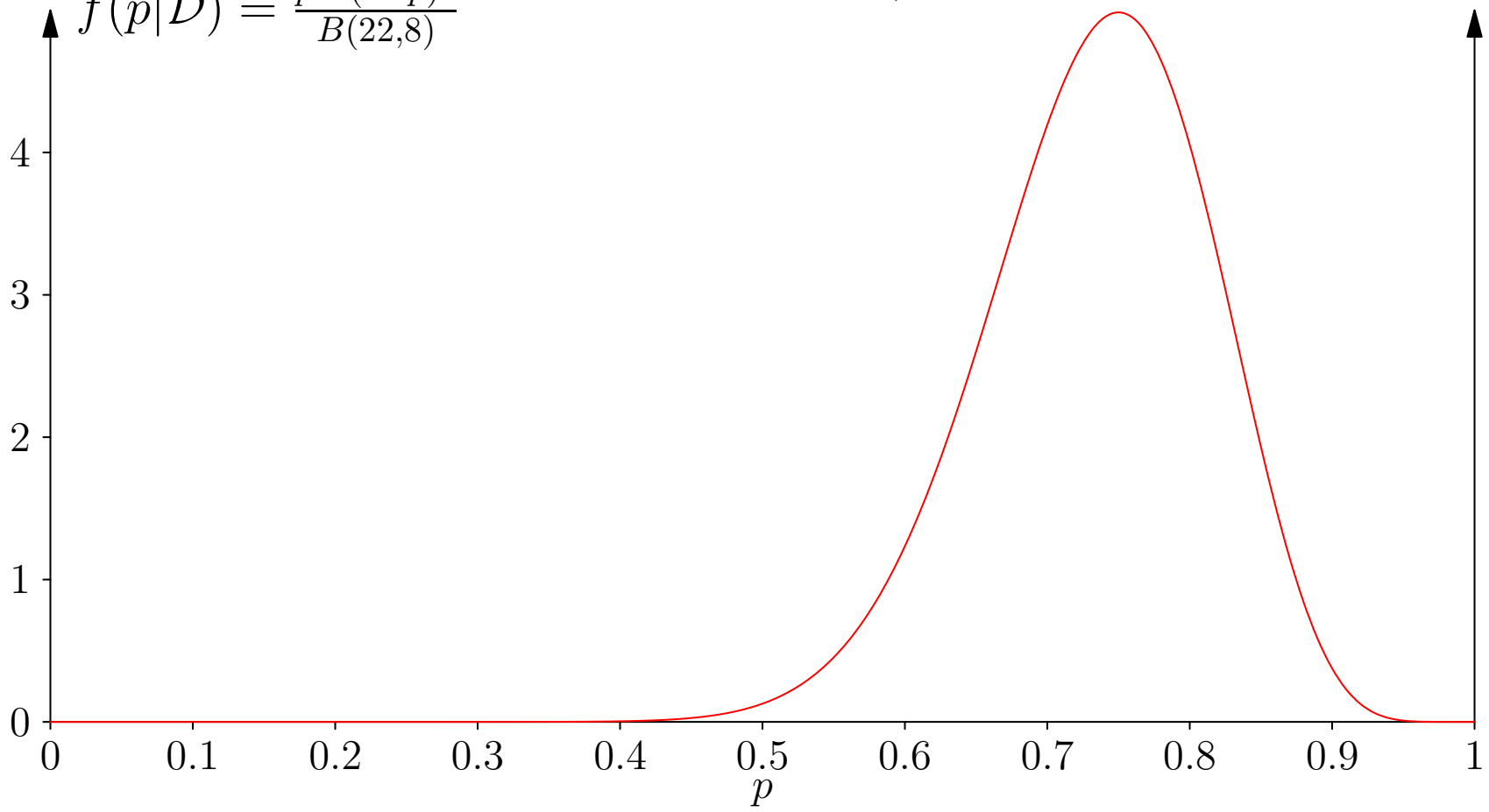
$$f(p|\mathcal{D}) = \frac{p^{20}\,(1-p)^{7}}{B(21,8)} \qquad a = 21,\ b = 8$$



Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,H,H,H,T,H,H,}$$

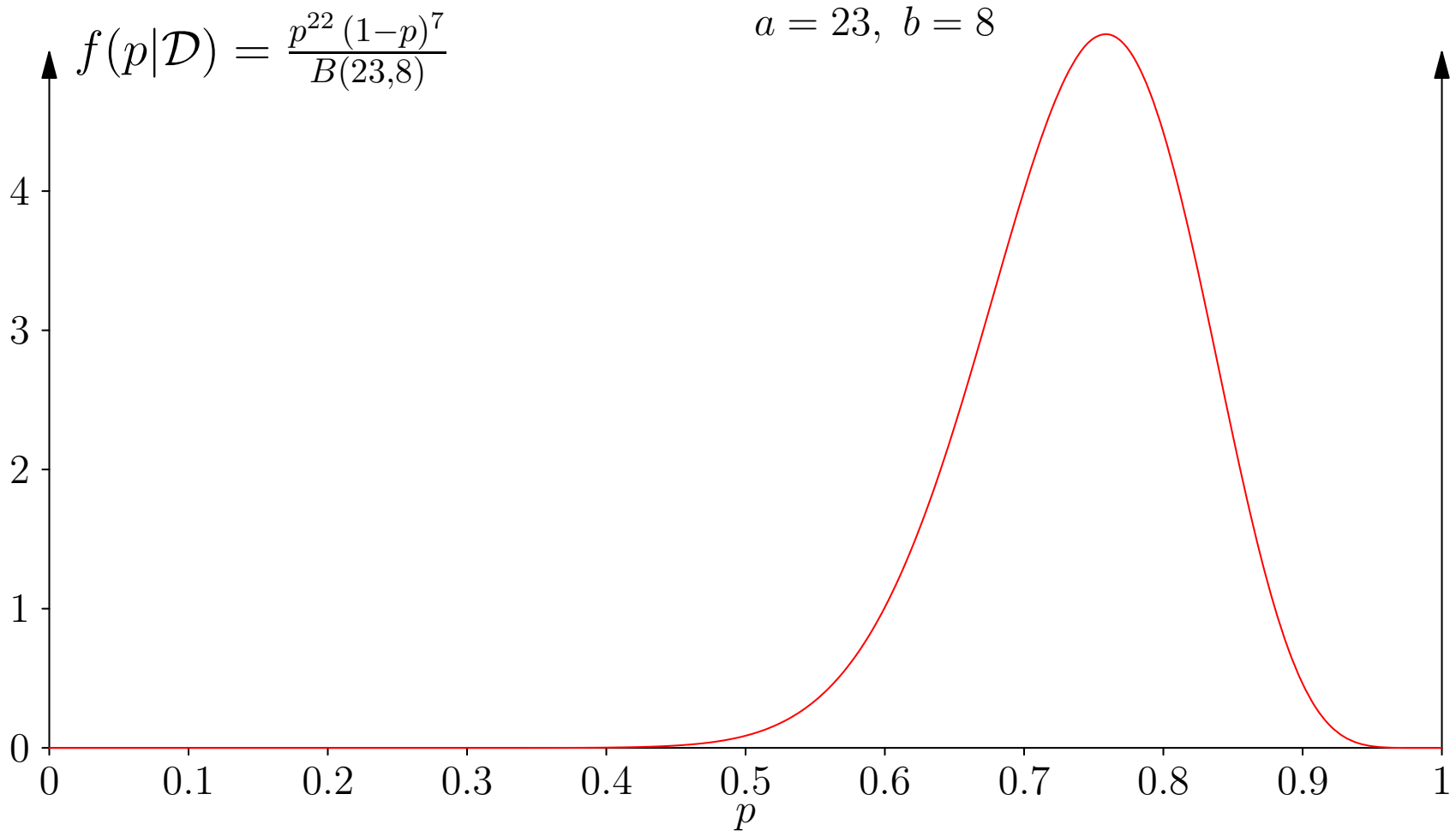
$$f(p|\mathcal{D}) = \frac{p^{21}\,(1-p)^{7}}{B(22,8)} \qquad a = 22,\ b = 8$$



Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,H,H,H,T,H,H,H,}$$

$$f(p|\mathcal{D}) = \frac{p^{22}\,(1-p)^{7}}{B(23,8)} \qquad a = 23,\ b = 8$$

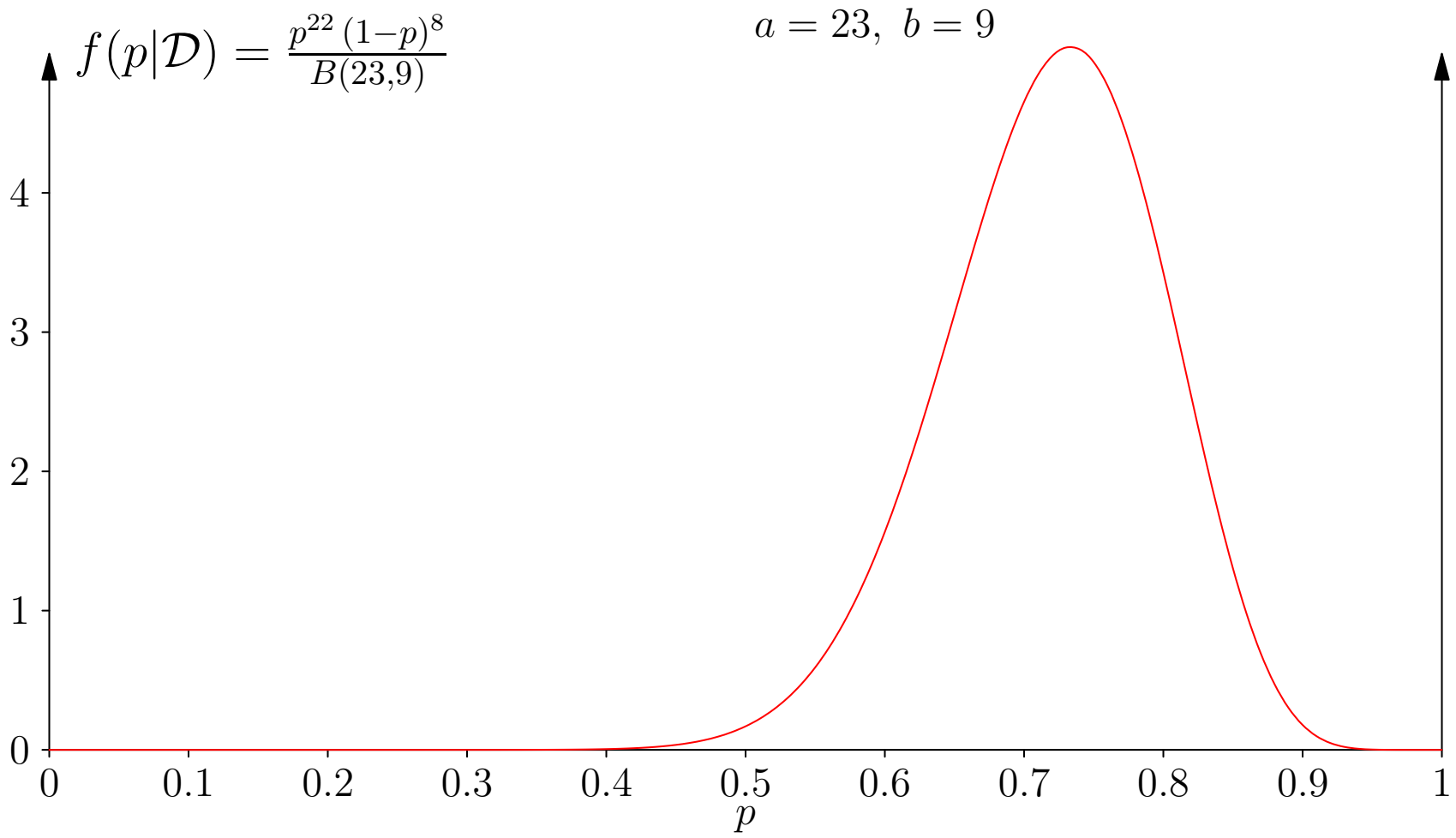


Example (p=0.7)

$$\mathcal{D} = \{\text{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,H,H,H,T,H,H,H,T}\}$$

$$f(p|\mathcal{D}) = \frac{p^{22}\,(1-p)^{8}}{B(23,9)}$$

$$a = 23,\ b = 9$$



Estimating Prediction Errors

- A full Bayesian treatment gives a prediction of its own error
- Assuming $f(p|\mathcal{D}) = \mathrm{Beta}(p|a,b)$ (in the example $a = 23$, $b = 9$ after 30 trials)
- The expected value of p is given by $a/(a+b) = 23/32 = 0.72$
- The standard deviation is

$$\sqrt{\frac{ab}{(a+b)^2(a+b+1)}} = 0.08$$

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Poisson Likelihoods

- Let's look at a second example of conjugate priors
- Suppose we want to find the rate of traffic along a road between 1:00pm and 2:00pm
- We assume the number of cars is given by a Poisson distribution

$$\mathbb{P}(N) = \mathrm{Pois}(N|\mu) = \frac{\mu^N}{N!}\mathrm{e}^{-\mu}$$

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Using Bayes

- Let us assume a Gamma distributed prior

$$p(\mu) = \mathrm{Gamma}(\mu|a_0,b_0) = \frac{b_0^{a_0}\mu^{a_0-1}\mathrm{e}^{-b_0\mu}}{\Gamma(a_0)}$$

- We will assume that we know nothing. The uninformative prior is $a_0 = b_0 = 0$
- The data is $\mathcal{D} = \{N_1, N_2, \ldots, N_n\}$
- The likelihood is $\mathrm{Pois}(N_i|\mu)$

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Posterior

- The posterior after seeing the first piece of data is

$$\begin{aligned} p(\mu|N_1) &\propto \mathbb{P}(N_1|\mu)p(\mu) \\ &\propto \frac{\mu^{N_1}}{N_1!}\mathrm{e}^{-\mu}\mu^{a_0-1}\mathrm{e}^{-b_0\mu} \\ &\propto \mu^{N_1+a_0-1}\mathrm{e}^{-(b_0+1)\mu} \end{aligned}$$

- The posterior is also a Gamma distribution $\mathrm{Gamma}(\mu|a_1,b_1)$ with $a_1 = a_0 + N_1$, $b_1 = b_0 + 1$

Posterior

- The posterior after seeing the first piece of data is

$$
\begin{aligned}
p(\mu|N_1) &\propto \mathbb{P}(N_1|\mu)p(\mu) \\
&\propto \frac{\mu^{N_1}}{N_1!}\mathrm{e}^{-\mu}\mu^{a_0-1}\mathrm{e}^{-b_0\mu} \\
&\propto \mu^{N_1+a_0-1}\mathrm{e}^{-(b_0+1)\mu}
\end{aligned}
$$

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Posterior

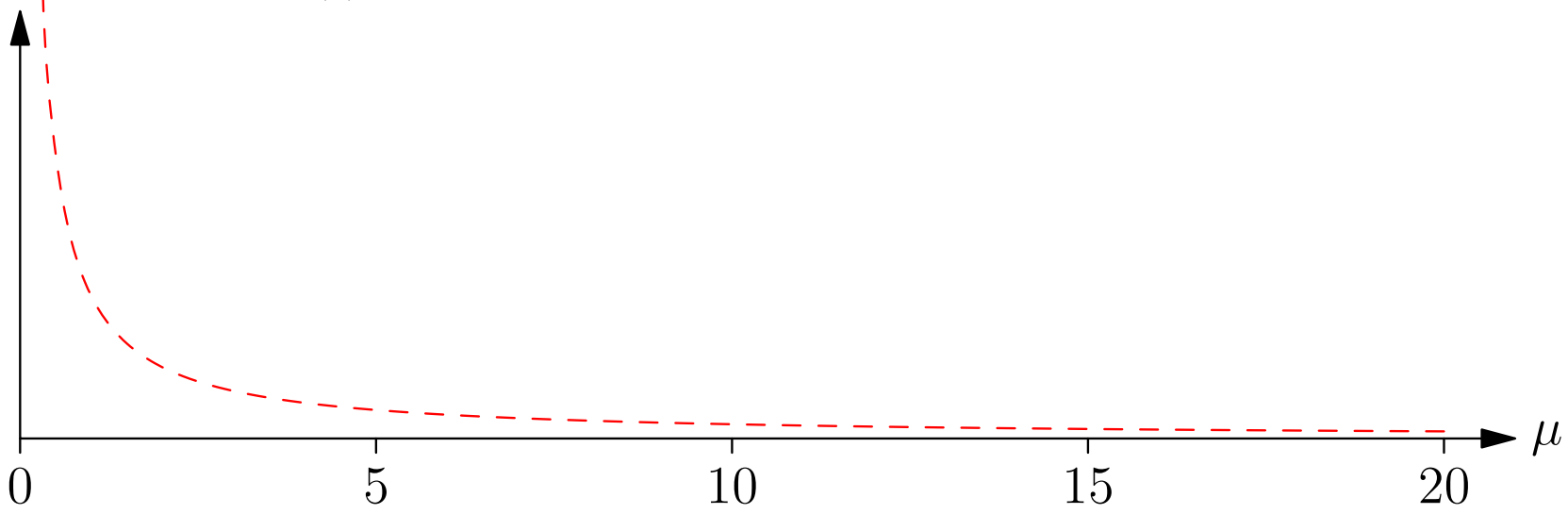
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Example ($\mu = 5$)

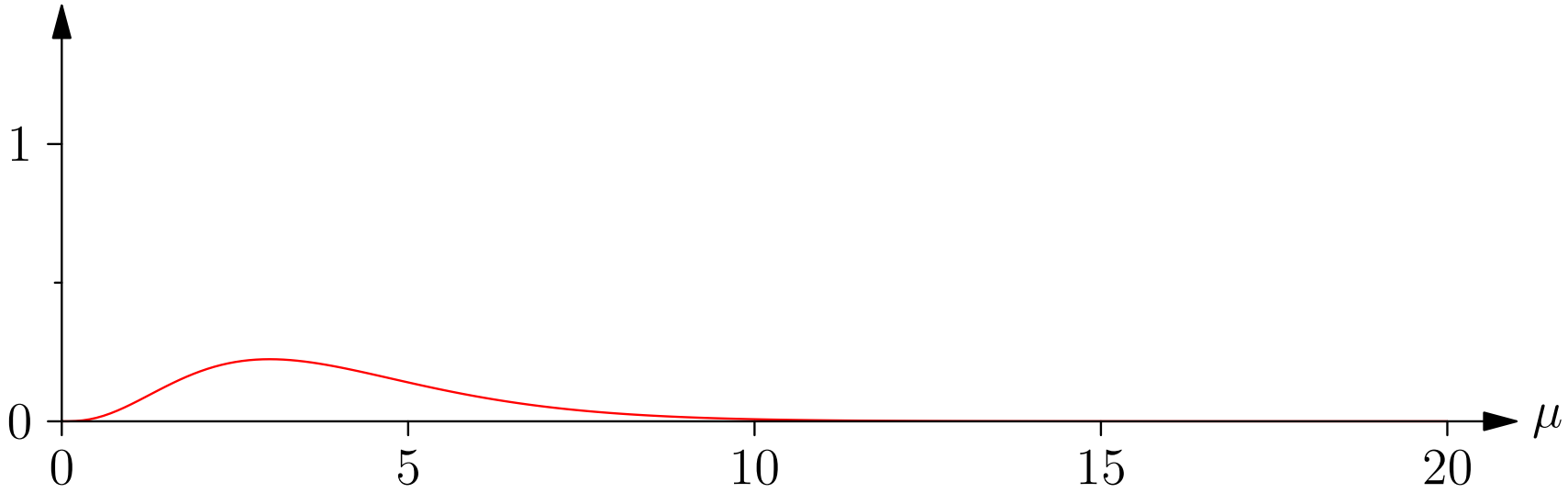
$$p(\mu|0,0) = \frac{0^0\,\mu^{-1}\,\mathrm{e}^{-0\,\mu}}{\Gamma(0)} \qquad a = 0,\ b = 0$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4\}$$

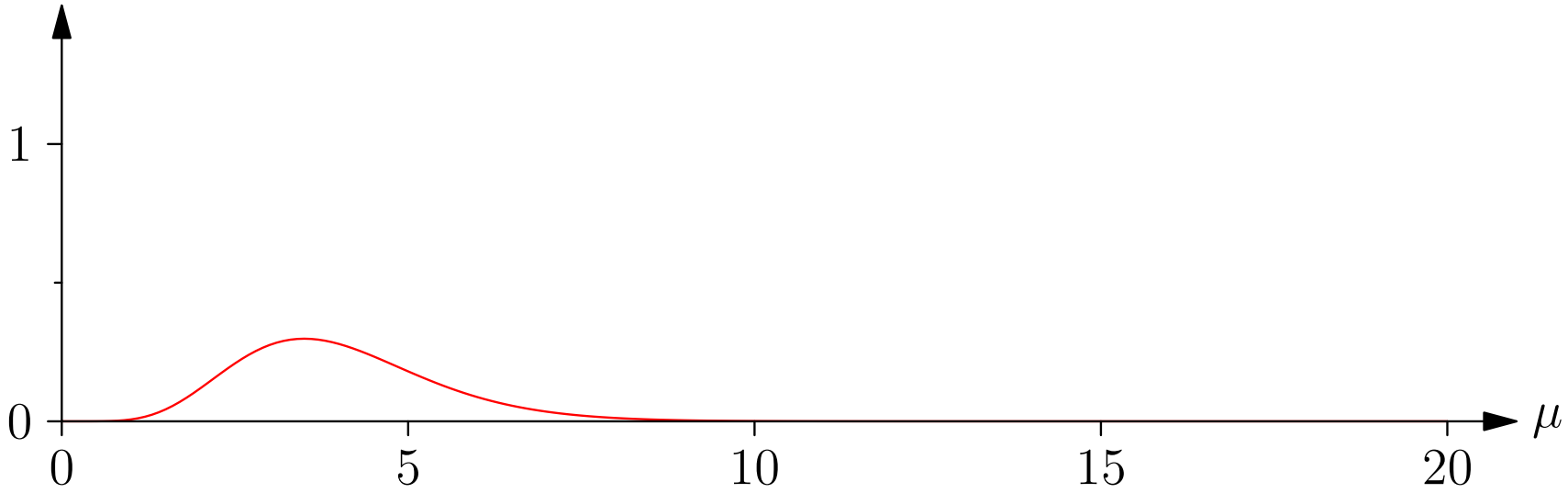
$$p(\mu|4,1) = \frac{1^4\,\mu^3\,\mathrm{e}^{-1\,\mu}}{\Gamma(4)} \qquad a = 4,\ b = 1$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4\}$$

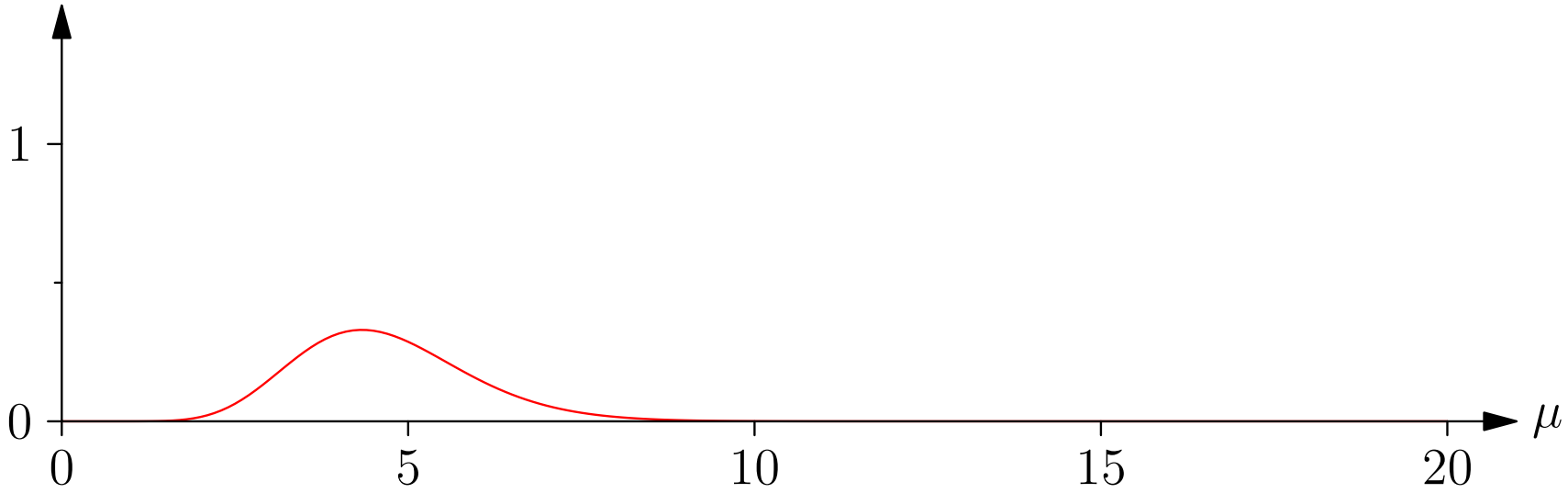
$$p(\mu|8, 2) = \frac{2^8\, \mu^7\, \mathrm{e}^{-2\,\mu}}{\Gamma(8)} \qquad a = 8,\ b = 2$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6\}$$

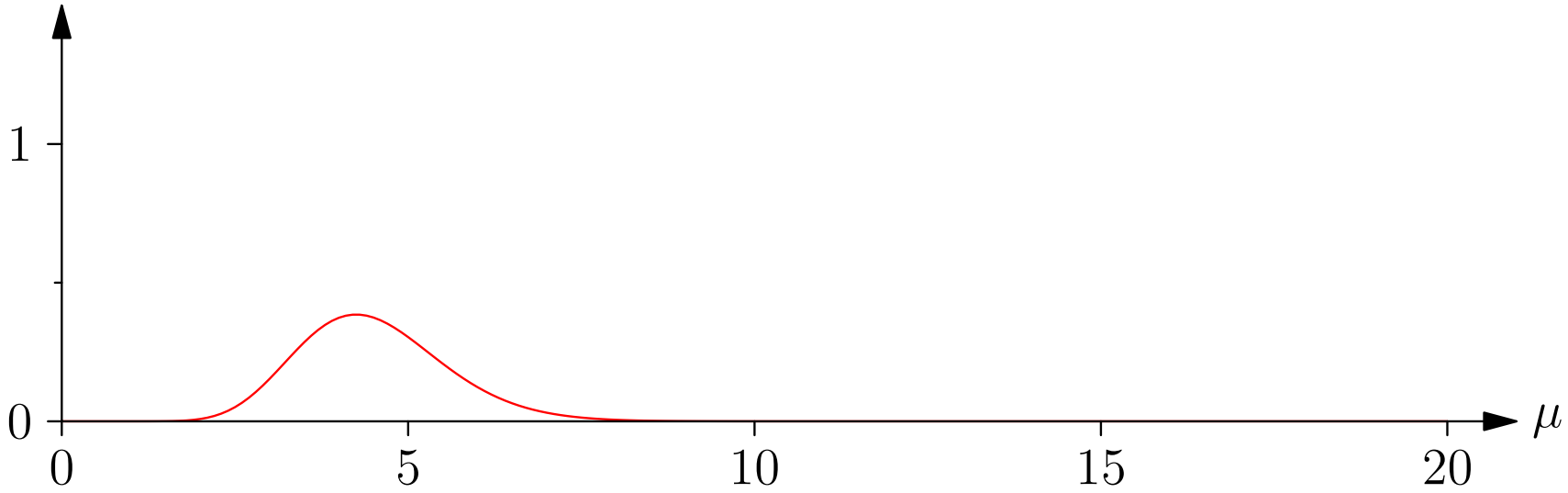
$$p(\mu|14, 3) = \frac{3^{14}\,\mu^{13}\,\mathrm{e}^{-3\,\mu}}{\Gamma(14)} \qquad a = 14,\ b = 3$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4\}$$

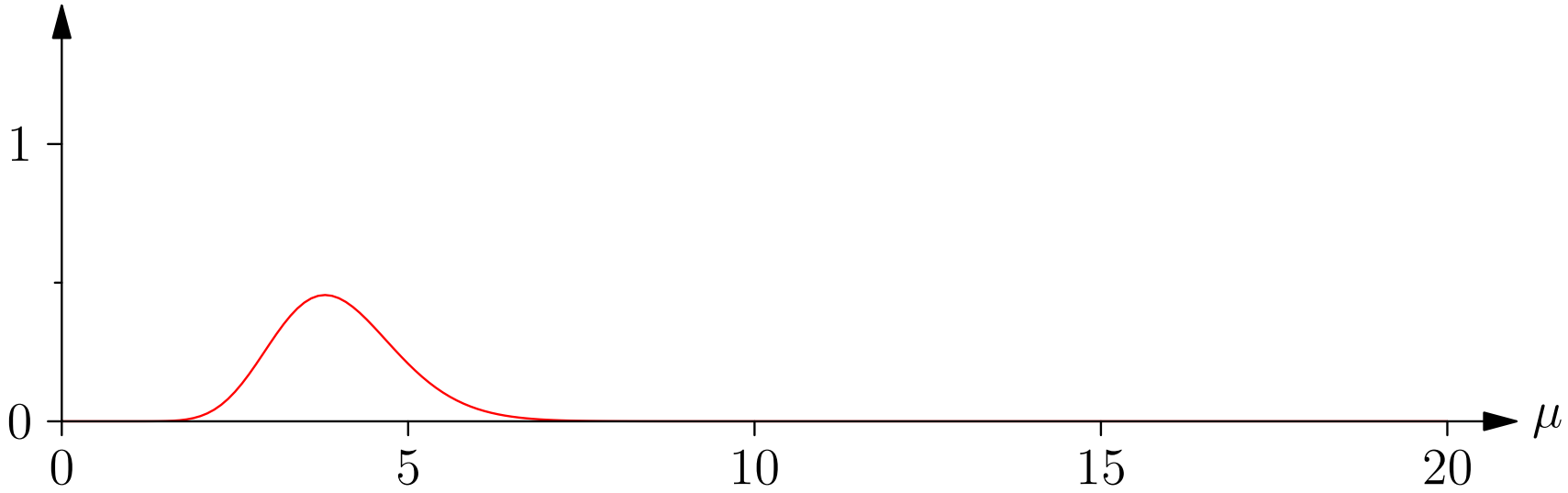
$$p(\mu|18, 4) = \frac{4^{18}\,\mu^{17}\,\mathrm{e}^{-4\,\mu}}{\Gamma(18)} \qquad a = 18,\ b = 4$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2\}$$

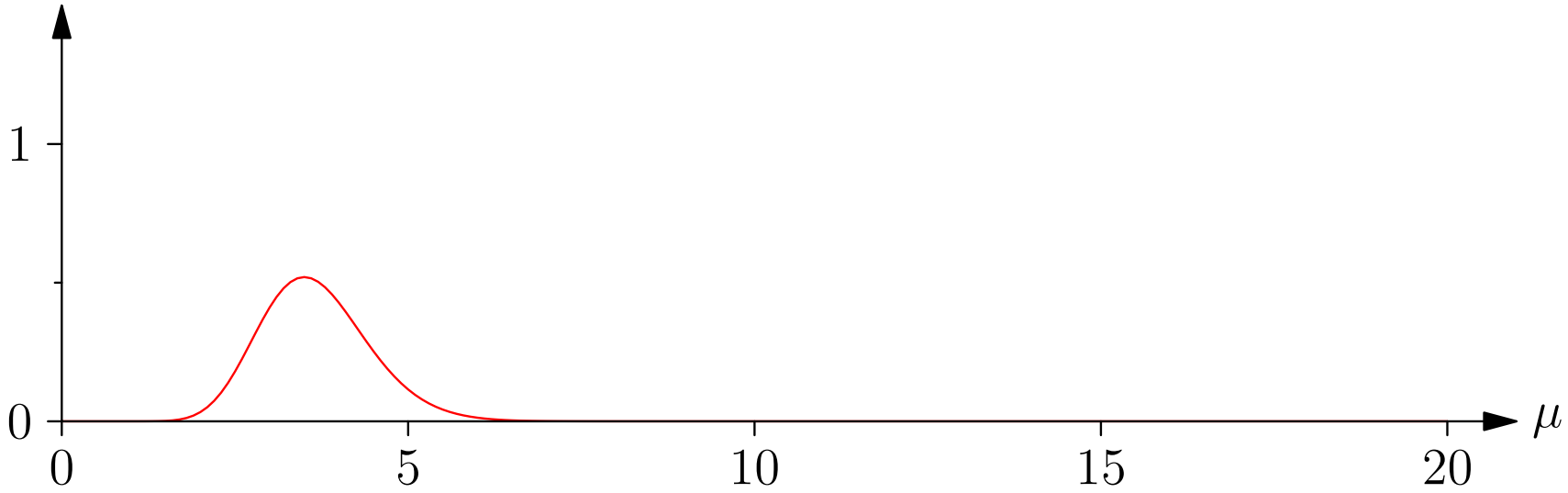
$$p(\mu|20, 5) = \frac{5^{20}\,\mu^{19}\,\mathrm{e}^{-5\,\mu}}{\Gamma(20)} \qquad a = 20,\ b = 5$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2\}$$

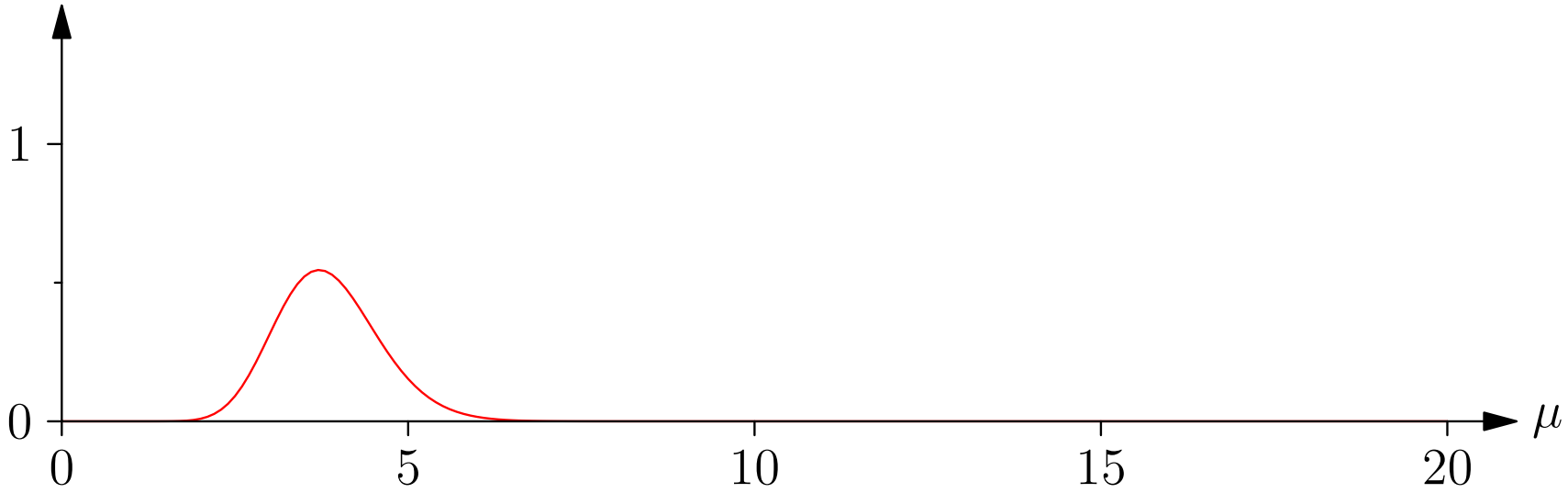
$$p(\mu|22, 6) = \frac{6^{22}\,\mu^{21}\,\mathrm{e}^{-6\,\mu}}{\Gamma(22)} \qquad a = 22,\ b = 6$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5\}$$

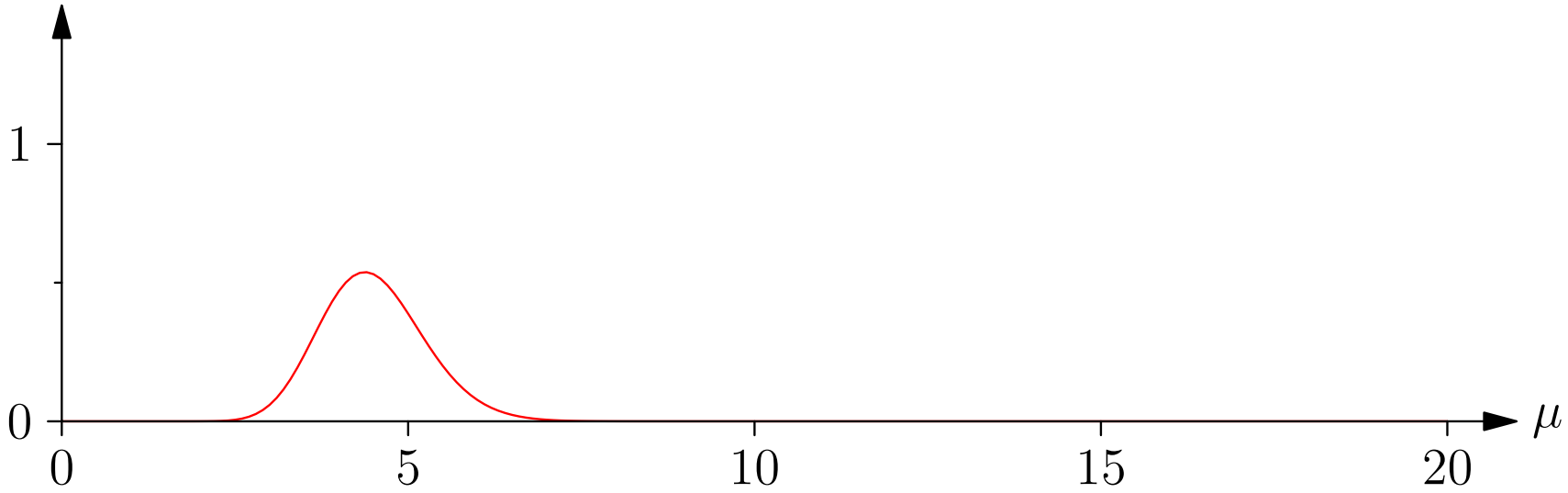
$$p(\mu|27, 7) = \frac{7^{27}\,\mu^{26}\,\mathrm{e}^{-7\,\mu}}{\Gamma(27)} \qquad a = 27,\ b = 7$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9\}$$

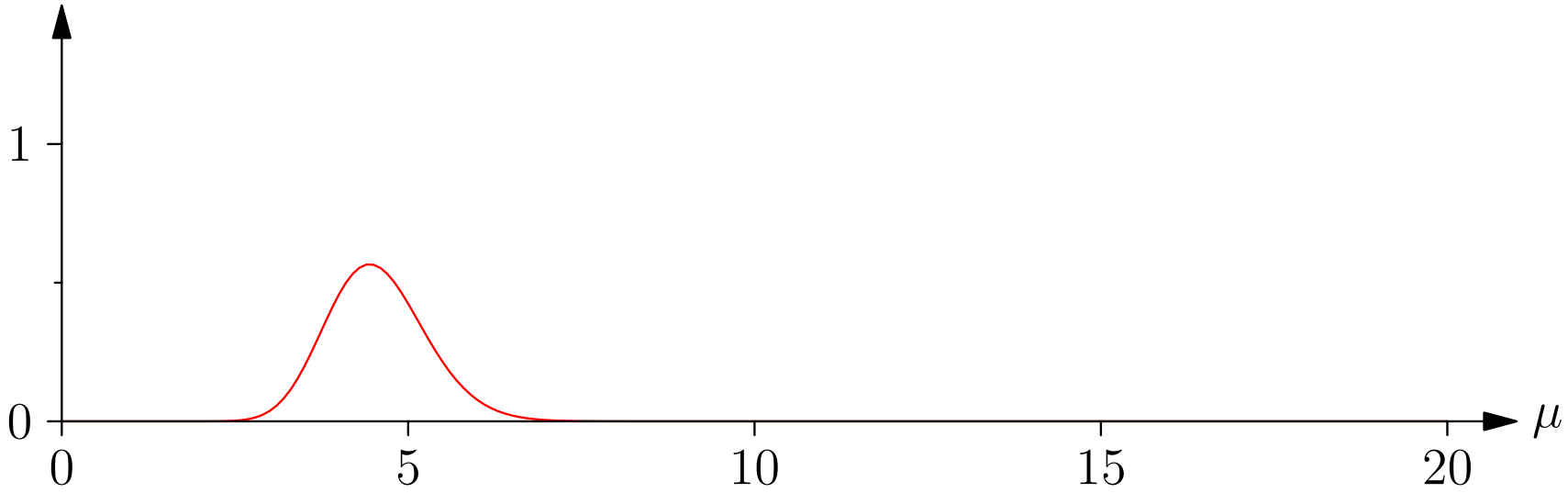
$$p(\mu|36, 8) = \frac{8^{36}\,\mu^{35}\,\mathrm{e}^{-8\,\mu}}{\Gamma(36)} \qquad a = 36,\ b = 8$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5\}$$

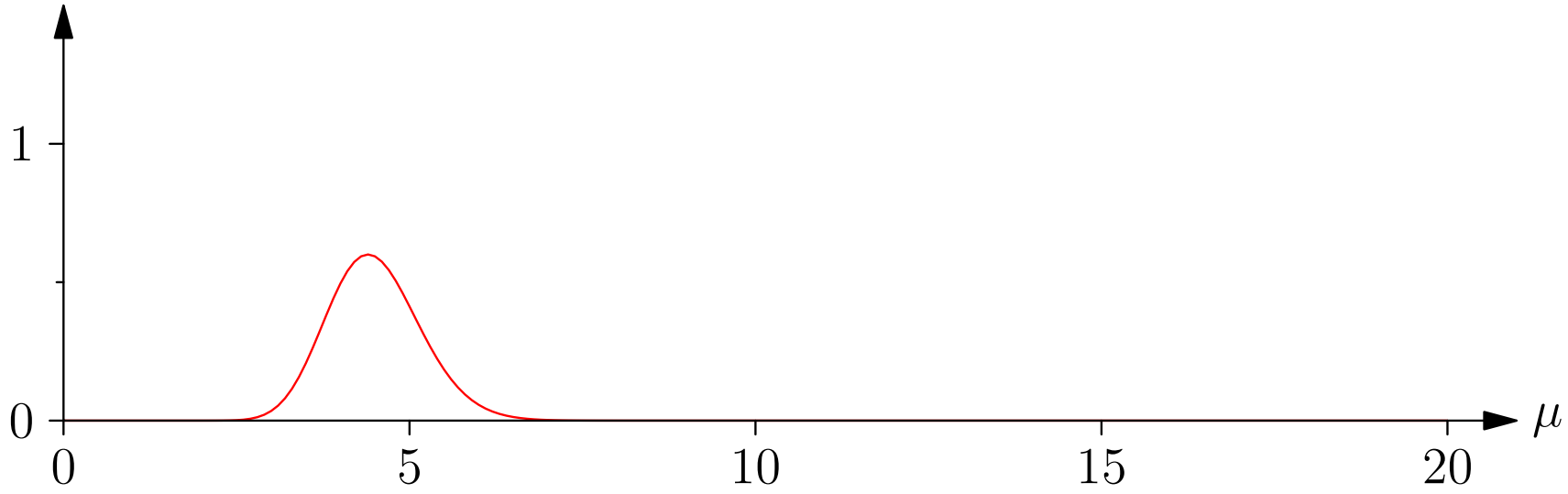
$$p(\mu|41, 9) = \frac{9^{41}\,\mu^{40}\,\mathrm{e}^{-9\,\mu}}{\Gamma(41)} \qquad a = 41,\ b = 9$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4\}$$

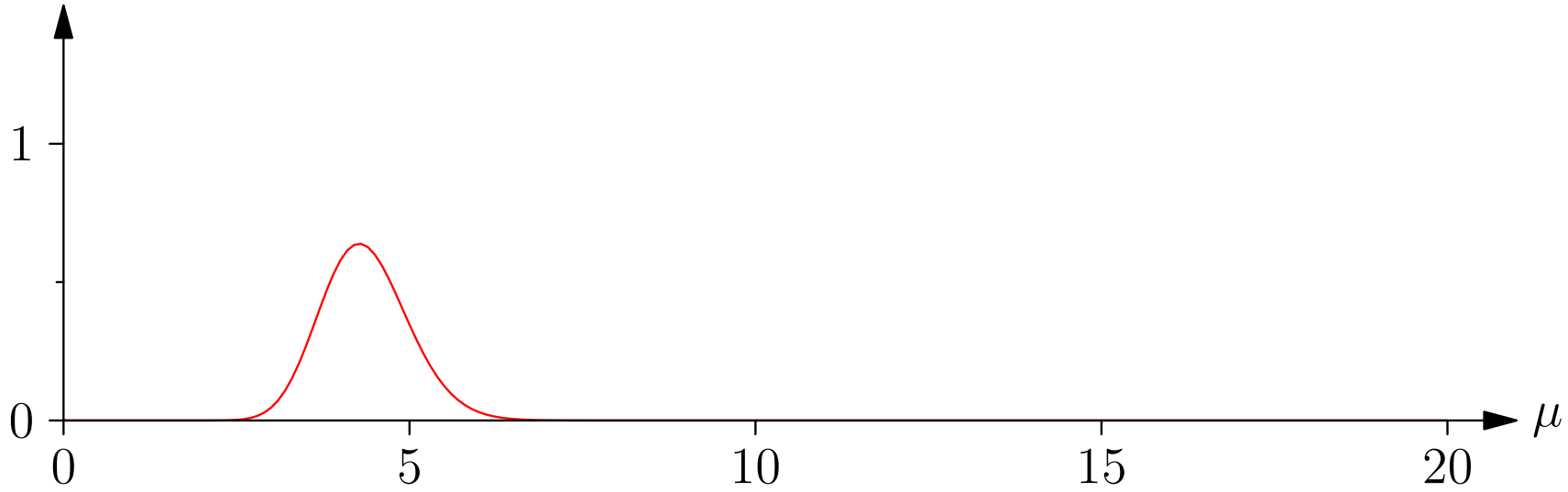
$$p(\mu|45, 10) = \frac{10^{45}\,\mu^{44}\,\mathrm{e}^{-10\,\mu}}{\Gamma(45)} \qquad a = 45,\ b = 10$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3\}$$

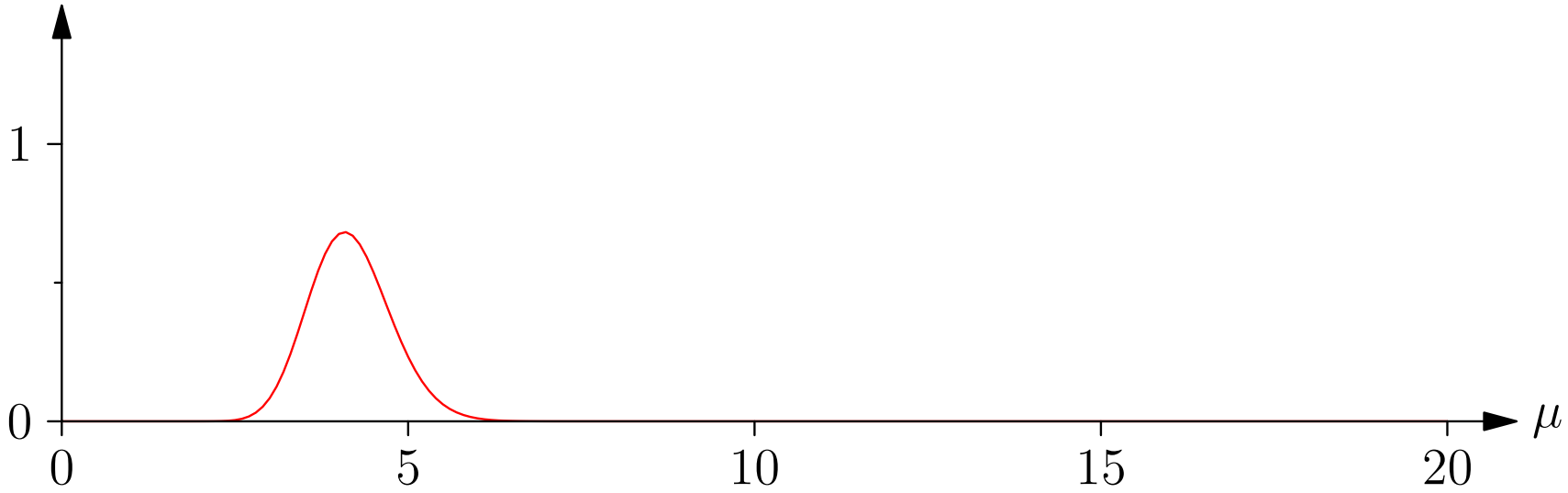
$$p(\mu|48, 11) = \frac{11^{48}\,\mu^{47}\,\mathrm{e}^{-11\,\mu}}{\Gamma(48)} \qquad a = 48,\ b = 11$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2\}$$

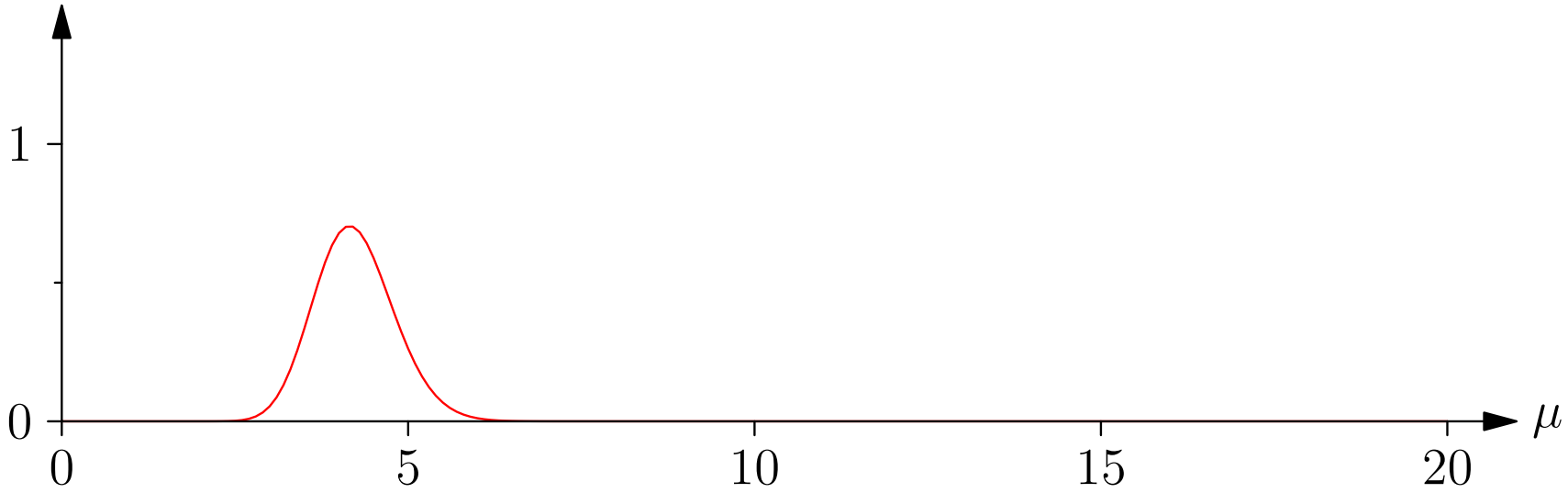
$$p(\mu|50, 12) = \frac{12^{50}\,\mu^{49}\,\mathrm{e}^{-12\,\mu}}{\Gamma(50)} \qquad a = 50,\ b = 12$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5\}$$

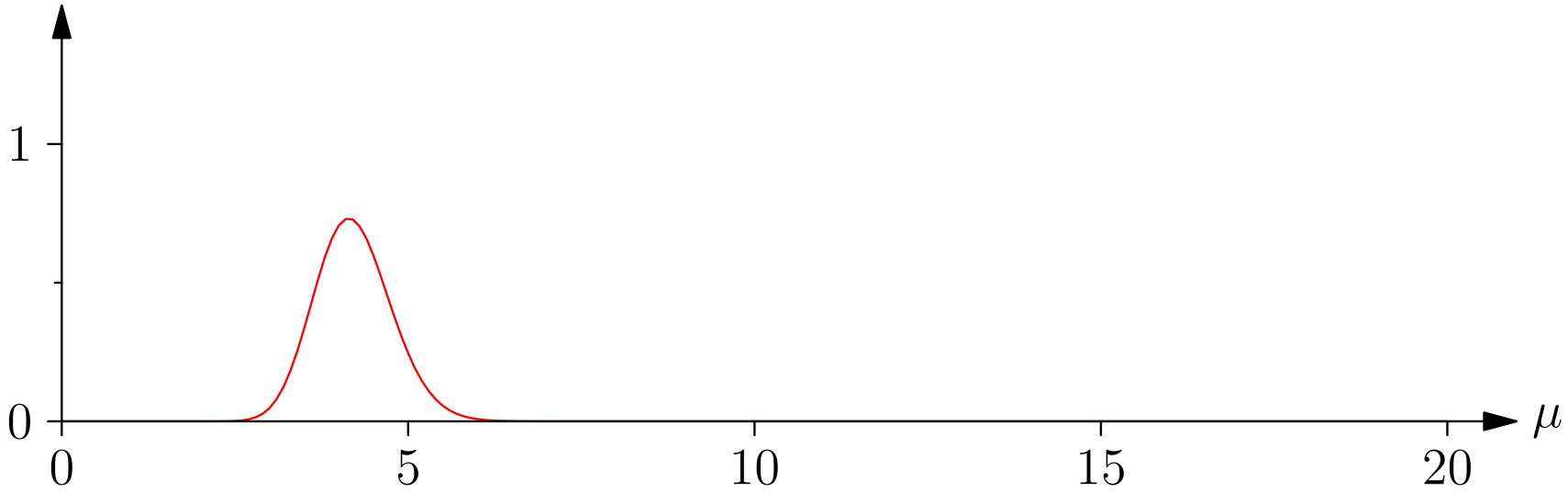
$$p(\mu|55, 13) = \frac{13^{55}\,\mu^{54}\,\mathrm{e}^{-13\,\mu}}{\Gamma(55)} \qquad a = 55,\ b = 13$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4\}$$

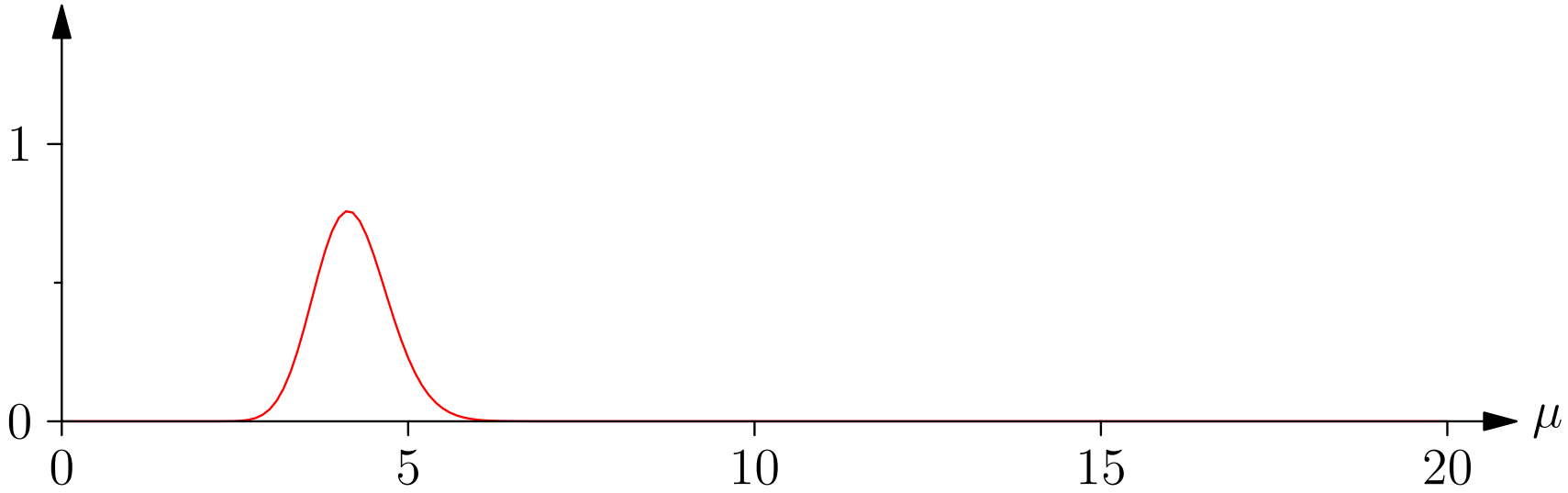
$$p(\mu|59, 14) = \frac{14^{59}\,\mu^{58}\,\mathrm{e}^{-14\,\mu}}{\Gamma(59)} \qquad a = 59,\ b = 14$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4\}$$

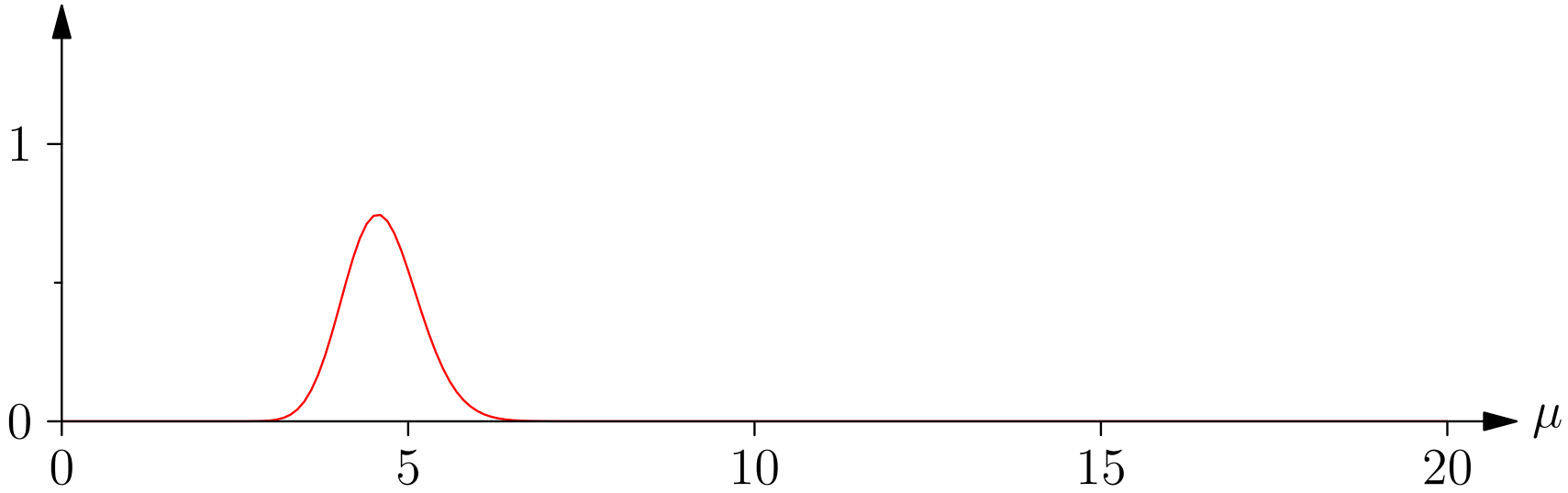
$$p(\mu|63, 15) = \frac{15^{63}\,\mu^{62}\,\mathrm{e}^{-15\,\mu}}{\Gamma(63)} \qquad a = 63,\ b = 15$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11\}$$

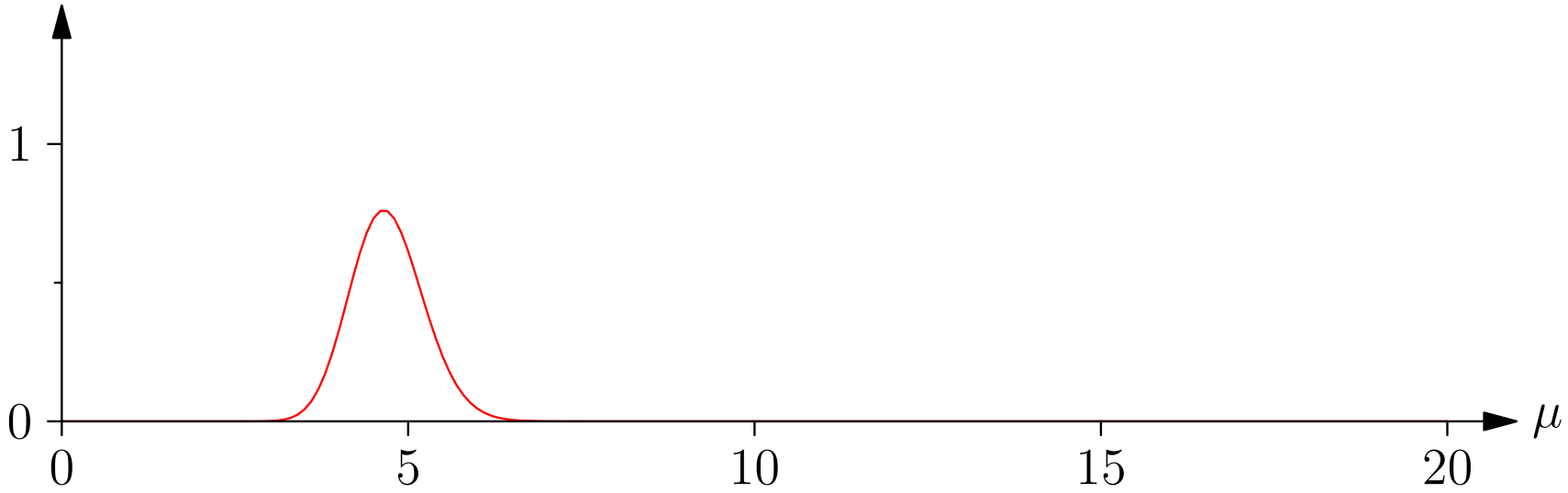
$$p(\mu|74, 16) = \frac{16^{74}\,\mu^{73}\,\mathrm{e}^{-16\,\mu}}{\Gamma(74)} \qquad a = 74,\ b = 16$$



Example ($\mu = 5$)

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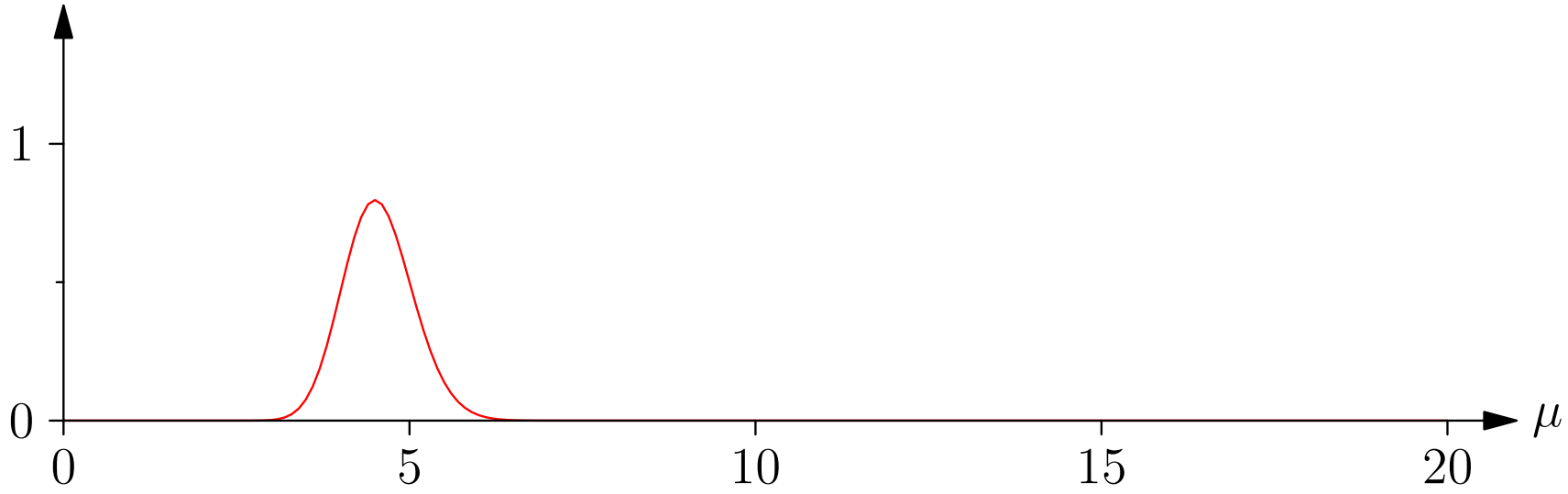
$$p(\mu|80, 17) = \frac{17^{80}\,\mu^{79}\,\mathrm{e}^{-17\,\mu}}{\Gamma(80)} \qquad a = 80,\ b = 17$$



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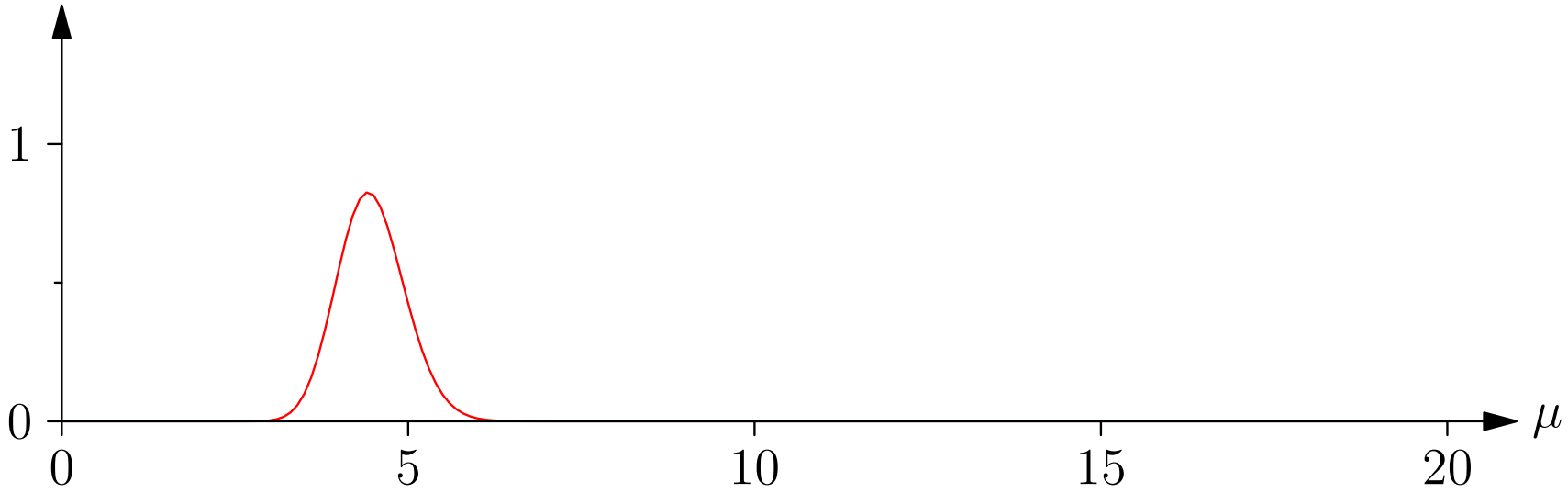
$$p(\mu|82, 18) = \frac{18^{82}\,\mu^{81}\,\mathrm{e}^{-18\,\mu}}{\Gamma(82)} \qquad a = 82,\ b = 18$$



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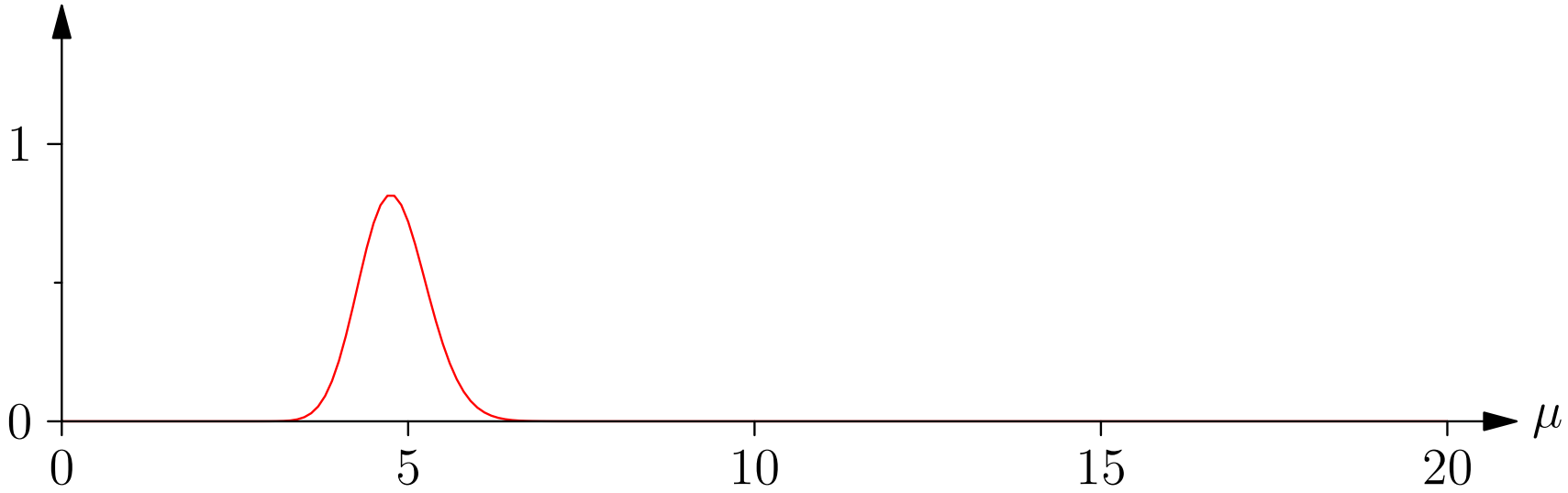
$$p(\mu|85, 19) = \frac{19^{85}\,\mu^{84}\,\mathrm{e}^{-19\,\mu}}{\Gamma(85)} \qquad a = 85,\ b = 19$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11, 6, 2, 3, 11\}$$

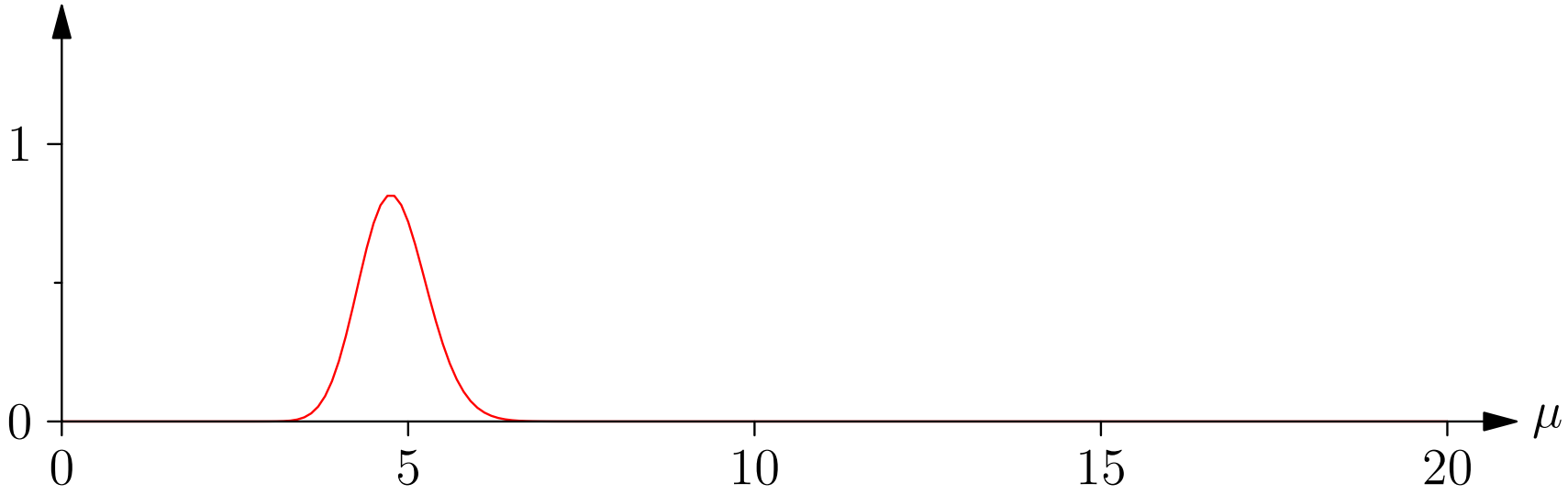
$$p(\mu|96, 20) = \frac{20^{96}\,\mu^{95}\,\mathrm{e}^{-20\,\mu}}{\Gamma(96)} \qquad a = 96,\ b = 20$$



Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11, 6, 2, 3, 11\}$$

$$p(\mu|96, 20) = \frac{20^{96}\,\mu^{95}\,\mathrm{e}^{-20\,\mu}}{\Gamma(96)} \qquad a = 96,\ b = 20$$



$$\mathbb{E}[\mu] = \frac{a}{b} = \frac{96}{20} = 4.8 \qquad \sqrt{\mathbb{V}\mathrm{ar}(\mu)} = \sqrt{\frac{a}{b^2}} = 0.5$$

Outline

1. Bayes' Rule
2. Conjugate Priors
3. **Uninformative Priors**



Uninformative Priors

- What if we have no prior knowledge, what should we do?
- OK usually we know whether we should make a measurement using a micrometer, ruler or car mileage, but we might still know almost nothing
- This led to Bayesian statistics being labelled as $subjective$
- However E. T. Jaynes (the greatest proponent of Bayesian methods) argued that we could answer this using symmetry arguments

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Uninformative Priors for Scale Parameter

- Why did we choose $a_0 = b_0 = 0$ implying a prior $p(\mu) = 1/\mu$?



- That is, we have no idea on what scale to measure μ

$$\int_A^B p(\mu)\mathrm{d}\mu = \int_{A/c}^{B/c} p(\mu)\mathrm{d}\mu$$

- Or $p(\mu) = \frac{1}{c}p(\frac{\mu}{c})$ implying $p(\mu) \propto \frac{1}{\mu}$

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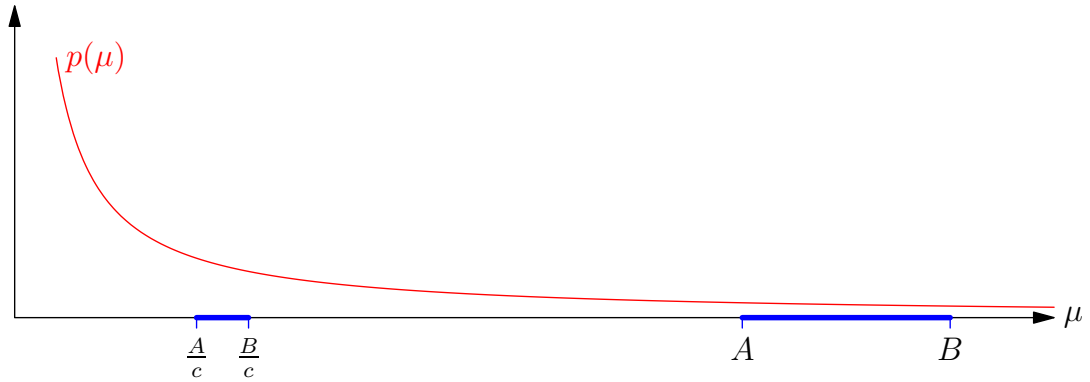
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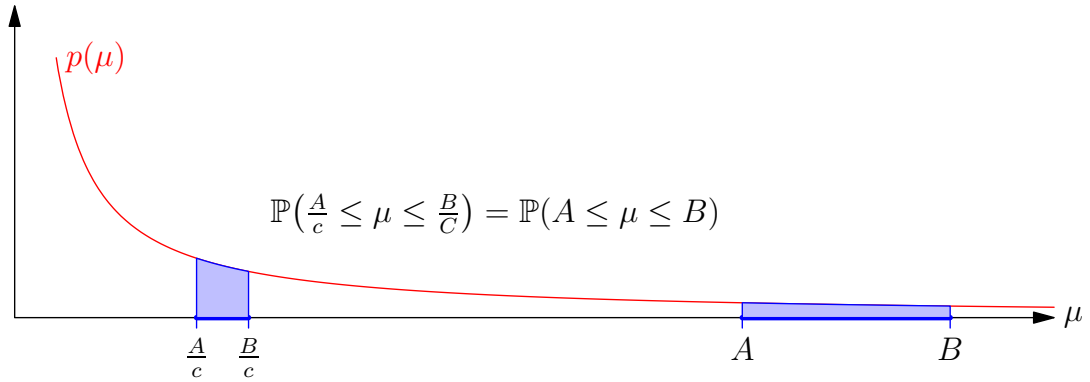
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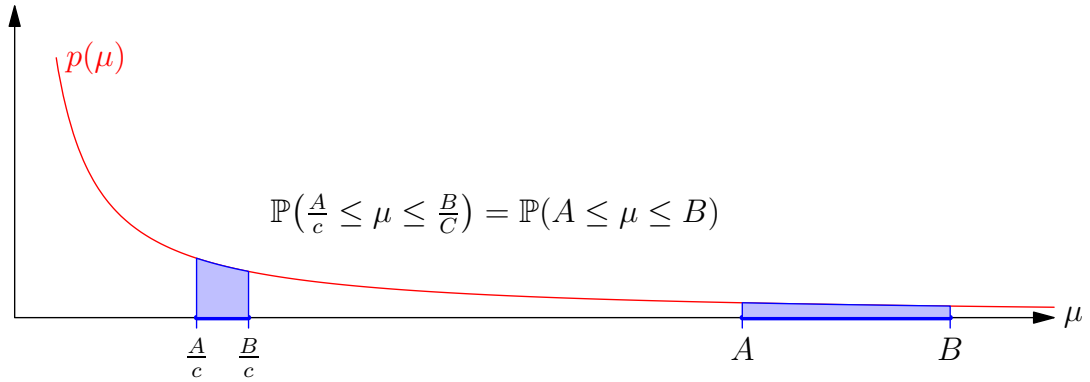
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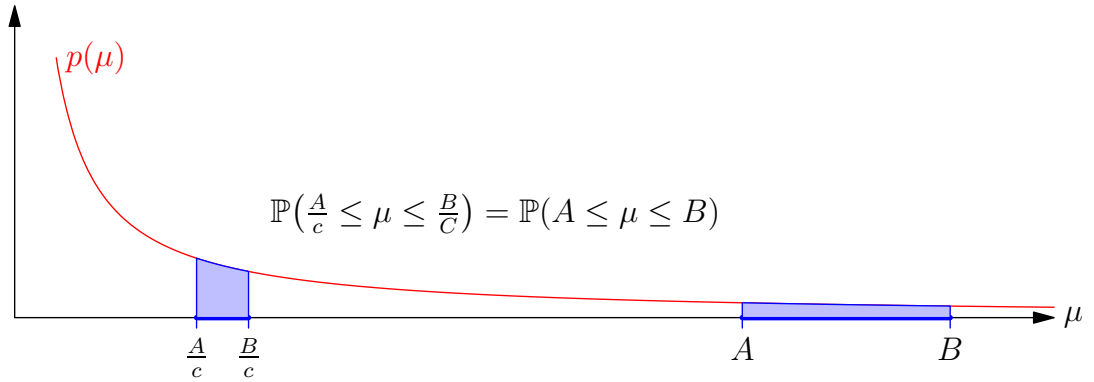
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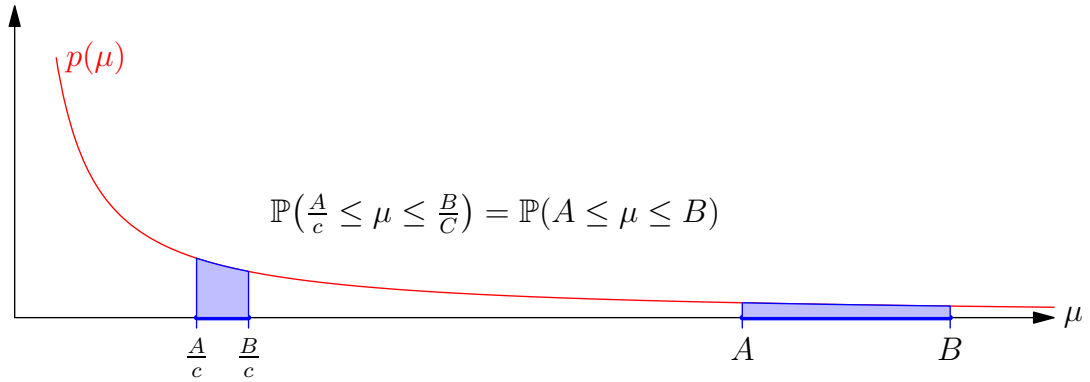
$$\int_A^B p(\mu)\mathrm{d}\mu = \int_{A/c}^{B/c} p(\mu)\mathrm{d}\mu = \int_A^B \frac{1}{c} p(\frac{\nu}{c})\mathrm{d}\nu$$

making a change of variables $\mu = \nu/c$

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Benford's Law

- Numbers occurring in life (physical constants, amounts of money) should not depend on the units (scale) measuring them
- They should then be distributed as $p(x) \propto 1/x$
- A curious consequence of this is that the first significant digit has a distribution

$$\mathbb{P}(\text{most s.f. of } x = n) = \frac{\int_n^{n+1} \frac{1}{x} \mathrm{d}x}{\int_1^{10} \frac{1}{x} \mathrm{d}x}$$

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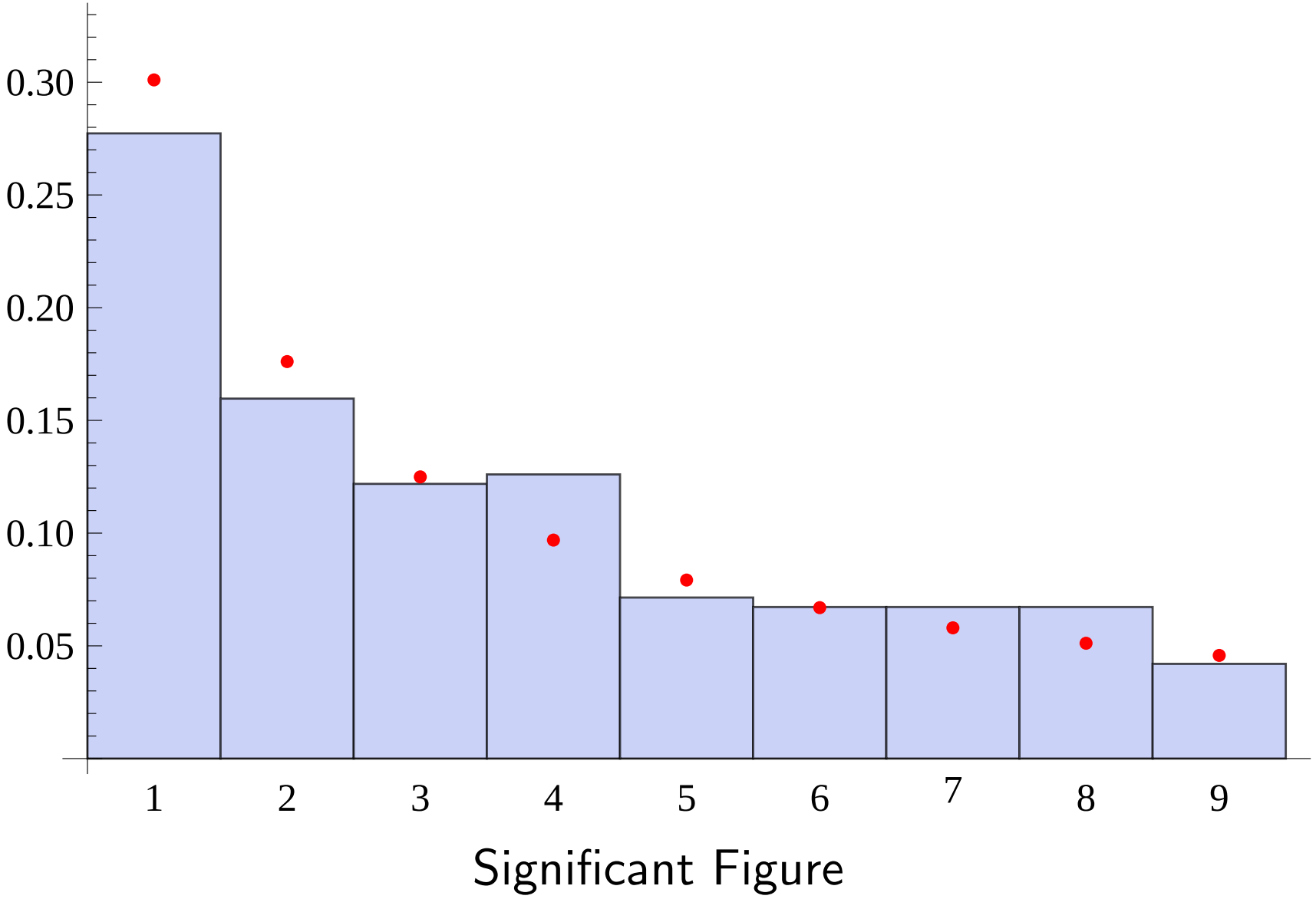
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Population Size of 238 Countries



Conclusion

- Bayesian inference provides a coherent framework which we can use for machine learning
- However, it requires a model of what is happening
- In practice Bayesian methods are easy if the data is generated from a likelihood with a conjugate prior distribution—we have to be clever to choose the right prior
- We will see later that much more frequently we will have likelihoods with no conjugate prior and we have to work much harder
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